MATHS in Journeys

Karl Jacklin

G.A.T.E.WAYS Publications

Quality ready-to-use resources

INSTANT LESSONS

MATHS in Journeys

Karl Jacklin

This book was commissioned by Jennifer Gough, project management was by Piper Print Media and Gail McManus Graphics designed the cover.

Karl Jacklin has been a teacher, school principal, Manager of Personnel and Administration in the North West Metropolitan Region of the Victorian Education Department, a teachers' college lecturer, tutor in statistics and measurement at Deakin University, and manager of his own software development and computer consultancy company. He has also presented many highly successful programs with G.A.T.E.WAYS (Gifted and Talented Education, Extension and Enrichment), an independent, non-funded organisation which aims to provide opportunities for children of 'like-minds' to participate in programs that will challenge them intellectually, and help develop their individual talents.

For further information: info@gateways.edu.au

G.A.T.E.WAYS directors, Win Smith and Jill Lawrence, wish to express their deep appreciation to Karl for his dedication in writing this exciting and challenging publication.

© G.A.T.E.WAYS Publications and Blake Education 2010
Blake Education owns all copyrights in the literary and the artistic works in this book.
Maps pages 4, 38, 39, 40, 41 and 42 © Melway Publishing Pty Ltd 2009
Maps pages 43 and 44 © Sydway Publishing Pty Ltd 2009

Copying of this book for educational purposes
The purchasing educational institution may only photocopy pages within this book in accordance with The Australian Copyright Act 1968 (the Act) and provided the educational institution (or body that administers it) has given a remuneration notice to the Copyright Agency Limited (CAL) under the Act. It is a breach of copyright to scan or in any other way make digital copies of the worksheets.

It is mandatory that ALL photocopies are recorded by the institution for CAL survey purposes.

For details of the CAL licence for educational institutions contact:
Copyright Agency Limited
Level 15, 233 Castlereagh Street
Sydney NSW 2000
Telephone: (02) 9394 7600
Facsimile: (02) 9394 7601
E-mail: enquiry@copyright.com.au

Printed in Australia by Ligare Book Printer

Blake EDUCATION
Better ways to learn

Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
info@blake.com.au
www.blake.com.au

ISBN 978 1 92571 019 9

PEFC™
PEFC/21-31-17

PEFC is committed to sustainable forest management through third party forest certification. For more information go to www.pefc.org

Introduction

Do you know people who are bewildered when faced with an urban street directory? ***Maths in Journeys*** introduces students to the skills they need to navigate street maps effectively.

Students will engage and extend their problem-solving skills as they complete mathematical tasks to follow directions for tracing a journey on the supplied real-life Melways and Sydways maps. These challenging tasks require students to follow instructions, recognise when they have made a wrong turn, and successfully backtrack using logical thinking and clues from the maps.

The first topic introduces students to some practical and mathematical aspects of working with maps. In the second topic, students extend on these skills and follow a step-by-step procedure for completing a journey. The journeys gradually increase in difficulty throughout the book.

To complete these tasks, students will need the map listed for each topic, a 30 cm ruler with millimetre markings, a set of compasses, a protractor and a calculator. If possible, enlarge the maps to A3 size. Photocopying the maps at different sizes does not affect the answers. However, note that it is essential for students to use the scale on the map they are actually working with in order to complete the journeys successfully.

Contents

1 **Mapping toolbox** **4**
No extra map needed

2 **Sink or swim!** **6**
Use Melways map 18

3 **Horses and courses** **10**
Use Melways map 28

4 **Monash mayhem** **12**
Use Melways map 70

5 **Cruising Boroondara** **14**
Use Melways map 45

6 **Sports and fast cars** **16**
Use Melways map 80

7 **Sporting jaunt** **18**
Use Melways map 28

8 **Medical mystery tour** **20**
Use Melways map 70

9 **Skating round schools** **22**
Use Melways map 18

10 **Off to the skate park** **24**
Use Melways map 45

11 **Dented enroute** **26**
Use Melways map 80

12 **Sydney sojourn** **28**
Use Sydways map 64

13 **From ABC to TV** **30**
Use Sydways map 35

14 **Traffic light trek** **32**
Use Melways map 28

15 **City circuit** **34**
Use Sydways map 64

16 **Racing round about** **36**
Use Melways map 80

Melways map 18 ..... **38**

Melways map 28 ..... **39**

Melways map 45 ..... **40**

Melways map 70 ..... **41**

Melways map 80 ..... **42**

Sydways map 35 ..... **43**

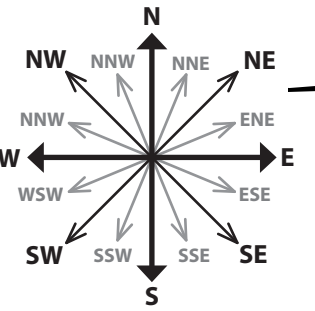
Sydways map 64 ..... **44**

Selected answers **45**

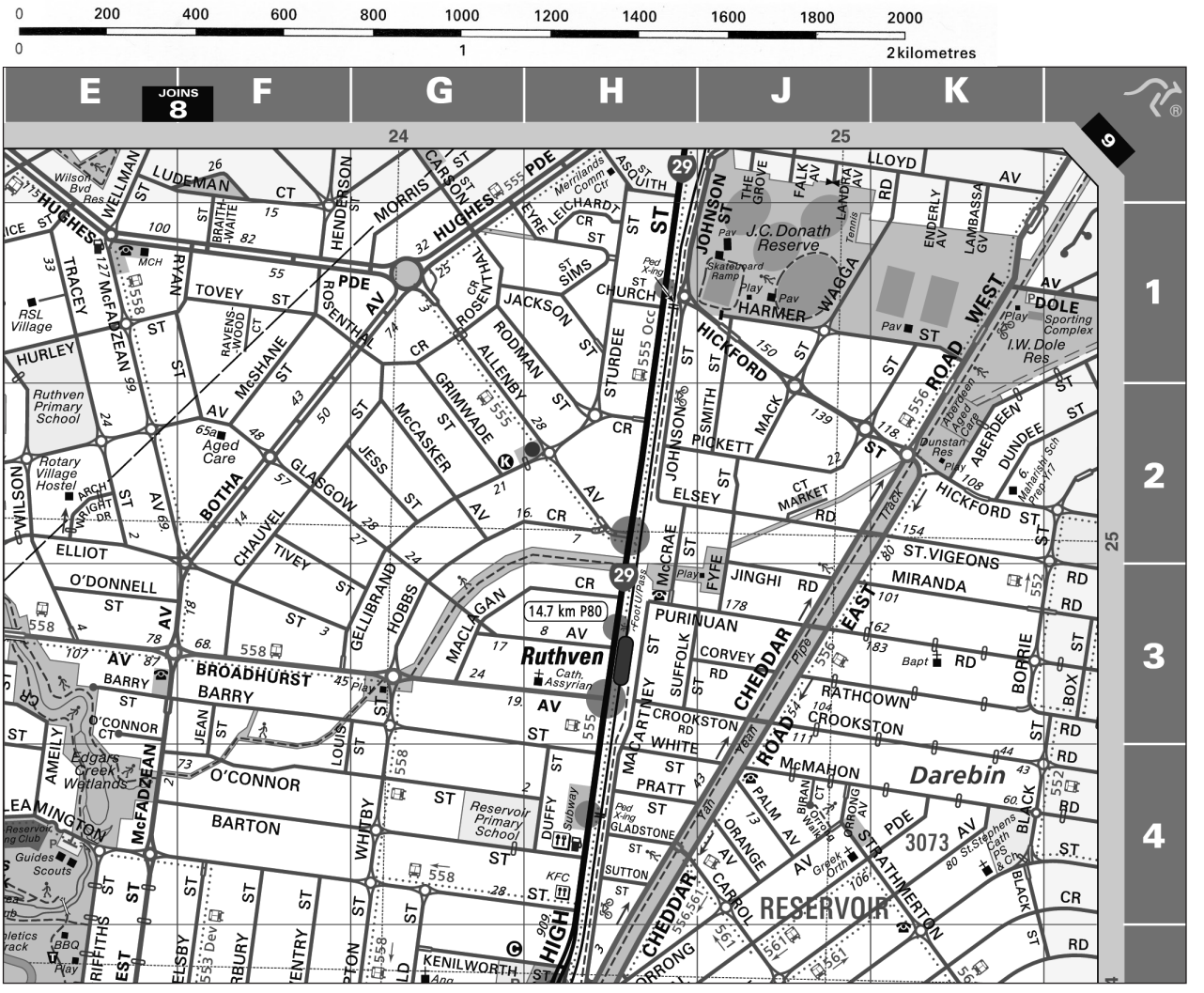
© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

1 Mapping toolbox



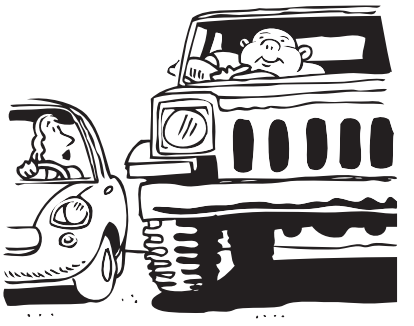
When navigating maps, the mathematics is about direction, scale, speed, distance and time. Through the activities below, you'll explore the features of maps and add some essential skills to your mathematics mapping toolbox.



Map copyright © Melway Publishing Pty Ltd 2009

Grid references

This street map uses grid references to help you locate features such as particular streets or railway stations. For example, Ruthven railway station has the grid reference H3. This means that it is located in column H, row 3.



1 *Circle Ruthven train station on the map.*

2 *Write the name of one street that passes through grid reference H3.*

3 *Which one of these streets does* **not** *pass through grid reference F2?*

a *Glasgow* **b** *Botha* **c** *McShane* **d** *Tovey*

4 *Chauvel Street starts in grid reference F3. Which other grid squares does it pass through?*

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

1 Mapping toolbox (cont'd)

Scale

5 *Look at the black and white rectangles above the map. What does this device tell you?*

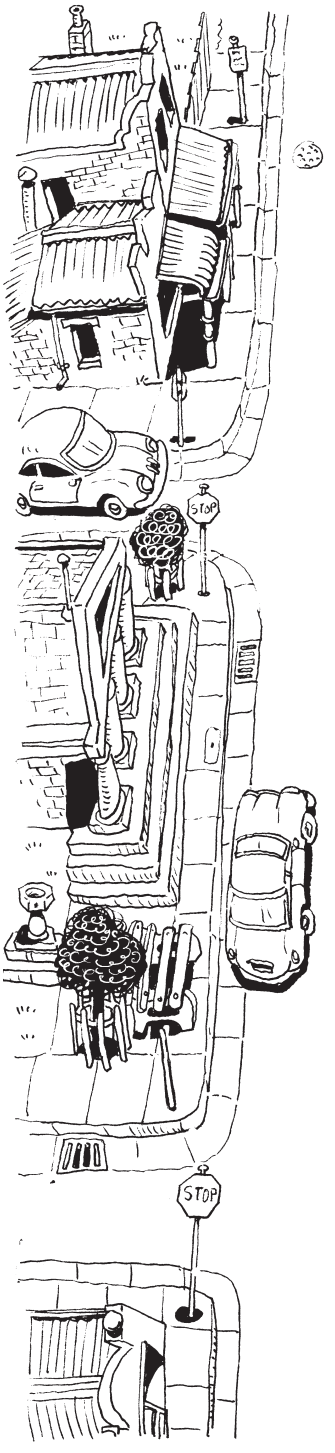
6 **a** *Find Barton Street in grid reference F4.*

b *Use your ruler to measure the distance from the intersection of Barton St and McFadzean St to the intersection of Barton St and Whitby St, to the nearest millimetre.*

c *Now measure the same distance along the scale above the map, from the point marked 0.*

d *According to the scale, approximately how many metres is that distance?*

7 *Follow the steps outlined above to work out the approximate length of High Street (column H).*



Distance

8 **a** *Find the intersection of Broadhurst Avenue and Gellibrand Street (Grid reference F3).*

b *Now find the intersection of Tracey Street and Hurley Street (Grid reference E1).*

c *What is the approximate distance between these two intersections in a* direct line*? (Don't forget to use the scale.)*

d *What is the distance between these two points via the streets on the map?*
(Hint: Use your ruler to draw the route on the map. Then measure each straight line length in centimetres. Add the lengths before you use the scale to work out the approximate distance.)

9 *Choose two more points on the map and challenge a classmate to work out the distance in a direct line as well as the distance via streets.*

Speed, time and distance

Remember:
speed = distance ÷ time

10 **a** *Find McFadzean Avenue (Grid reference E1).*

b *The length of McFadzean Avenue is 1 600 metres. If you drive the length of this avenue at an average speed of 10 metres/second, how many seconds will it take?*

11 **a** *Find Botha Avenue (Grid reference F2).*

b *If you drive along this avenue at an average speed of 10 metres/second for 40 seconds in total, how many metres will you travel?*

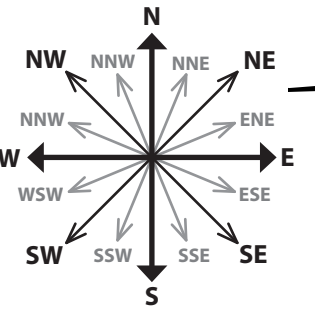
12 *Think of another question about speed, time and distance. Challenge a classmate to solve it.*

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

2 Sink or swim!

Melways map 18



Your task is to follow these directions on the given map to reach a final destination. This journey introduces you to some of the features of maps and calculations involving scale, distance, speed and time.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map scale

1 *Look at the scale on your map.*
It is a graphic scale and looks like a long rectangle divided into black and white sections of equal size.

Each small rectangle indicates a length on your map.
The measurements above the scale indicate the matching length in real life.

a *Measure the divisions and reproduce them below.*

b *Now label the scale above with the distances in real life indicated by each division. Include the units.*

2 *Use the scale you reproduced above to answer the following questions.*

a *What distance in real life does the length of one rectangle on your scale represent?*

b *What distance in real life does the length of four rectangles on your scale represent?*

c *What length on your map indicates 1 km in real life?*

d *What distance in real life does 2 cm on your map represent?*

3 *Complete the rule below.*

Distance in real life in metres ÷ _____ = length in centimetres on the map.

Grid references

These street maps use grid references to help you locate features such as particular streets or railway stations. For example, Preston railway station has the grid reference F11. This means that it is located in column F, row 11.

When you find it on your map, you will see that some of the station icon is in row 12. However, most of it is in row 11, so this is the grid reference given.

4 *Match each railway station to the correct grid reference.*

a *Ruthven*	• *G9*
b *Reservoir*	• *H3*
c *Regent*	• *G6*

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

2 Sink or swim! (cont'd)

The journey begins

Map symbols you need:

- roundabout
- public telephone
- railway station
- P car park

5 ***Begin at the map's centre point – the intersection of the lines between columns E and F, and rows 6 and 7.***

Your starting position is **1.6 km** ***in a direct line from the centre point.***

a ***Use your scale to work out what length on your map represents 1.6 km (1 600 m).***

b ***Set your compasses to the length you worked out above.***

c ***Use your compasses to draw a circle with this radius. Your starting point lies on this circle.***

Clue 1: Measuring clockwise from due north, it is on an angle of 231° from the centre point.

Clue 2: You won't need a photographic memory to work this out.

A Drive in a southerly direction on the main road adjacent to your starting position at an average speed of **60 km/h** for **44 s.**

- **You will travel 'south-ish' down E _ _ _ _ _ _ _ _ Street.**

  Hint: Streets do not follow compass directions exactly – you need to allow for this. Look out for words like 'southerly' and 'easterly' – this tells you the direction is not exactly south or not exactly east.

  - **What distance in metres would you travel at a speed of 60 km/h for 44 s?**

    Start by converting 60 km/h to a speed in m/s.

    60 km/h = ? m/s

    There are 60 × 1 000 m in 60 km

    60 000 m = 60 km

    1 hour = 60 min = 60 × 60 s = 3 600 s

    60 km/h = $\frac{60\,000}{3\,600}$ m/s = 16.67 m/s (rounded to 2 decimal places)

  - **At 16.67 m/s, how many metres will you travel in 44 s?**

    16.67 m × 44 s = _____

  - Using your answer to question 3 on the previous page, how many centimetres on your map is this? _____

  - You are now at the intersection of Elizabeth Street and _ _ _ _ _ _ Road.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Tip: *Draw your journey on the map to help you keep track of where you are.*

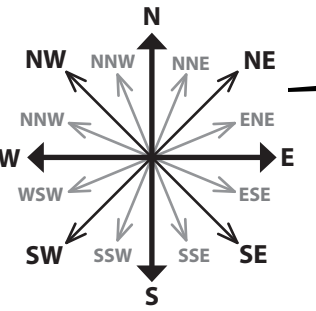
- Remember, you need to keep track of your speed, distance and driving time for each section of your journey so you can calculate the totals at the end. Write the distance in the table at the right.

Direction	*southerly*
Distance	
Driving time	*44 seconds*
Speed	*60 km/h, 16.67 m/s*

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

2 Sink or swim! (cont'd)



B Turn **left**. Unfortunately the traffic has become more congested and you can only average **30 km/h**. Drive at this speed for **3 min 46 s.**

- If you are travelling south and turn left, in which direction will you then be travelling?

  _____
- What speed in m/s is 30 km/h? _____
- At this speed, how many metres will you travel in 3 min 46 s? _____

  Hint: First convert 3 min 46 s to seconds:

  (3 × 60) + 46 = _____
- What length on your map is this in cm? _____
- You are now at the intersection of Murray Road and _ _ _ _ Street.
- Fill in the table at the right for this section.

Tip: *Use the compass picture above. Turn the page so that the south direction arrow is pointing directly in front of you. Which direction is to your left? You could also do this with the map.*

Direction
Distance
Driving time
Speed

C Turn **left**. The traffic is light again. Proceed in this direction for **2.37 km** at **60 km/h**.

- What length on your map is this in centimetres? _____
- You are now at the intersection of High Street and _ _ _ _ _ _ _ R _ _ _ W_ _ _.
- Fill in the table at the right for this section.

Direction
Distance
Driving time
Speed

D At the intersection, veer **21°** to the **right**. Proceed in this new direction for **1.63 km** at **60 km/h.**

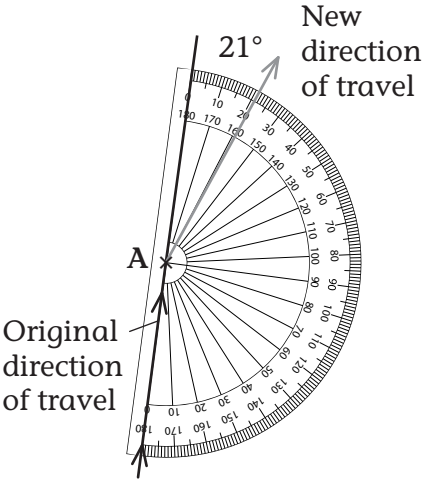
- To veer 21° to the right, you need to position your protractor carefully.

  Make sure the baseline of the protractor is positioned on the road on which you are travelling. Extend this baseline straight through the intersection to make it easier to check the position of your protractor.

  Place the point marked 'A' on the diagram at the right directly on your current location (centre of the intersection).

  Mark 21° on your map.
- You are now travelling along _ _ _ _ _ _ _ R _ _ _ W_ _ _.
- What length on your map in centimetres will you travel along this road?

  _____
- Fill in the table at the right for this section.



Direction
Distance
Driving time
Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

2 Sink or swim! (cont'd)

E Turn **left** at the roundabout. Proceed at **60 km/h** for **37.4 s**.

- How many metres will you travel in 37.4 s at 60 km/h? Remember, ***speed = distance ÷ time***.

  60 km/h = 16.67 m/s

  37.4 s × 16.67 m/s = _____

  - How many centimetres on the map is this distance? _____
  - Fill in the table at the right for this section.

Direction
Distance
Driving time
Speed

F You need to make an important telephone call – but you left your mobile at home. Proceed to the closest accessible public telephone. Unfortunately, road works restrict you to **40 km/h**. You will need to turn right at a T-intersection and proceed a short distance to the telephone box, which is located halfway to the next street corner.

Direction
Distance
Driving time
Speed

Hints:

- Use the map symbols on page 7 to find out how a public telephone is shown on the map.
- You will need to make four turns to reach the public telephone. In order, these turns are left, left, right, right.
- For each street you travel, list the name, the distance you drive and the time it takes. Record the totals for this section in the box above.

G Turn **left** at the corner and proceed down Macartney Street for **71.2 s** at **50 km/h**.

Direction	Distance
Travel time	Speed

H Turn **right** at the intersection. At **60 km/h**, drive on for another **1 min 22 s** to yet another roundabout.

Direction	Distance
Travel time	Speed

I Take the shortest route to your destination – The Preston Yacht Club – travelling at **60 km/h** to the Emma Avenue carpark **(Map 18 D5)**. List the streets and distances you drive and the time it takes.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

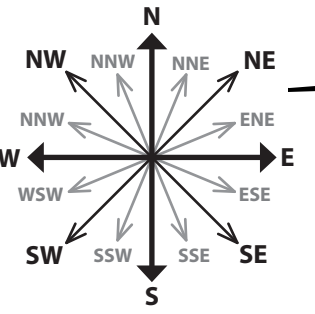
Hint: Average driving speed (km/h) = $\frac{\text{Total distance driven (km)}}{\text{Total driving time (h)}}$

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

3 Horses and courses

Melways map 28



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- roundabout
- **P** carpark
- **M** motel

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
Your starting position is* 1.96 km *in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 102° from the centre point.

Clue 2: It's horses for courses, and you're on top of a T-intersection.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A Drive in a **southerly** direction on the road adjacent to your starting position at an average speed of **50 km/h** for **39 s**.

Direction	Distance
Driving time	Speed

B Turn **right**. Unfortunately the traffic has become more congested and you can only average **40 km/h**.
After **31 s** driving at this speed, the traffic thins out and you pass across a major road.

Direction	Distance
Travel time	Speed

3 ***Name the major road.***

C After you cross the major road, increase your speed to **60 km/h**. Continue at this speed for **2 min 42 s**.

Direction	Distance
Travel time	Speed

D Turn **left**. The traffic continues to be light. Proceed on this carriageway for **almost 1 km** at **60 km/h**.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

3 Horses and courses (cont'd)

E At the roundabout, turn **right** and proceed in a **northerly** direction for **340 m** at **40 km/h.** Turn **left** into the fourth carpark you pass. Do not include this short distance into the carpark in your journey calculations.

Direction	Distance
Travel time	Speed

4 ***Which activity are you most likely to be doing from your current location?***

a ***Shopping***

b ***Going to the cinema to see a movie***

c ***Visiting your old school***

d ***Walking in the park***

F Leave the car park and return to the roundabout on the main road. Unfortunately you're caught in a slow procession of cars and can only manage this distance at **25 km/h**.

Direction	Distance
Travel time	Speed

G When you get to the roundabout, turn **left** and proceed for **1 min 11 s** at **50 km/h**.

Direction	Distance
Travel time	Speed

5 ***What is the name of the waterway on your right?***

6 ***What two aquatic groups are based at the northern end of the park on your right?***

H Turn **right** at the T-intersection and proceed for **820 m** at **60 km/h.**

Direction	Distance
Travel time	Speed

I Turn **right** again. Drive for **2.28 km** at **60 km/h** and park your car. You've reached your destination, a motel you are staying at tonight.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

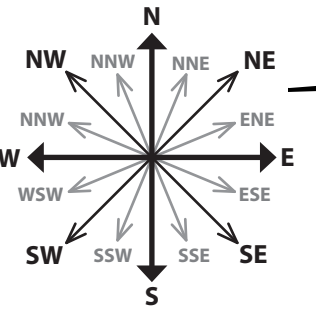
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

4 Monash mayhem

Melways map 70



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- roundabout
- P carpark
- restaurant

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
Your starting position is 2.11 km in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 173° from the centre point.

Clue 2: It's the main entry to an academic institution named after a famous Australian soldier.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A Drive in a **westerly** direction from your starting position at an average speed of **80 km/h** for **29 s**.

Direction	Distance
Travel time	Speed

3 ***What is the connection between your starting point and the name of the road on which you have just driven?***

B Veer **30° right**. Continue at the same speed for **25 s**.

Direction	Distance
Travel time	Speed

C Turn **right**. The traffic is light. Proceed on this carriageway for **3 km** at **70 km/h**.

Direction	Distance
Travel time	Speed

4 ***How many public telephones have you seen from the road since beginning your journey? Give the grid reference for each one.***

D At the intersection, turn **left** and proceed for **1.63 km** at **80 km/h**. Park your car near the intersection and have a look around.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

4 Monash mayhem (cont'd)

5 ***You're hungry – what is the closest place to buy food?***

E You've finished your meal so it's off again. Turn **left** at the traffic lights after waiting at the red light for **45 s**.

Direction	Distance
Travel time	Speed

F Drive for **410 m** at **70 km/h**.

Direction	Distance
Travel time	Speed

G Enter the freeway in an **easterly** direction. Proceed for **3 km** at **100 km/h**.

Direction	Distance
Travel time	Speed

H Slow to **80 km** as you enter the **390 m** off-ramp.

Direction	Distance
Travel time	Speed

I Turn **right**. Travel in a **southerly** direction for **1 min 55 s** at **70 km/h**.

Direction	Distance
Travel time	Speed

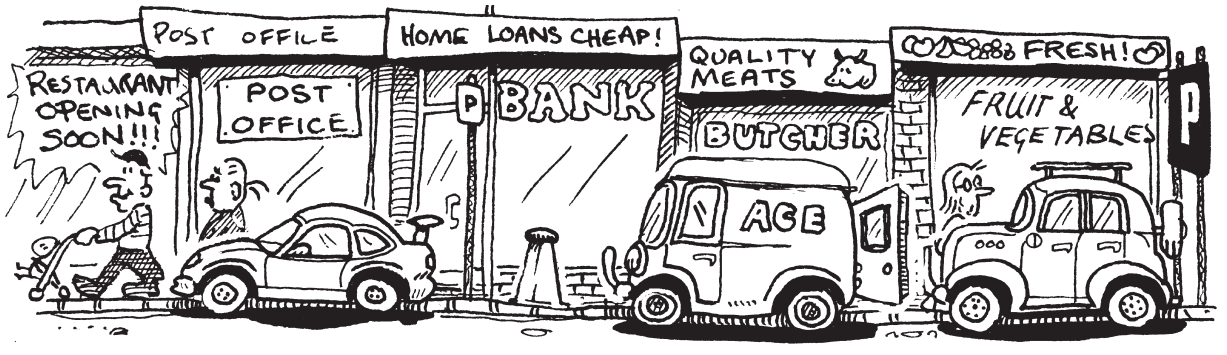
6 ***You pass a reserve on your right that contains a picturesque lake. What is the name of the reserve?***

J Turn **left** and drive **50 m** at **20 km/h** to the car park of your final destination.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

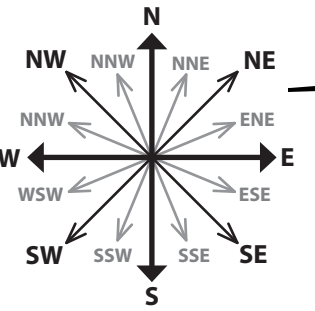


© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

5 Cruising Boroondara

Melways map 45



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- public telephone
- restaurant
- roundabout

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
Your starting position is 2.62 km in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 327° from the centre point.

Clue 2: An extra pair of eyes for a visually impaired person.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A Drive **240 m** to the nearest main road at **20 km/h**. Turn **left**.

Direction	Distance
Travel time	Speed

B Drive for **760 m** at **60 km/h**. Turn **right**.

Direction	Distance
Travel time	Speed

C Proceed at **60 km/h** for **1 min 36 s**. Turn **left**.

Direction	Distance
Travel time	Speed

D Proceed for **1.72 km**.

Direction	Distance
Travel time	Speed

3 ***Which activity are you most likely to do from your current location?***

a ***Play a game of croquet***

b ***Watch a game of cricket***

c ***Cook a barbecue***

d ***Visit the grave of a friend or relative***

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

5 Cruising Boroondara (cont'd)

E Proceed in the **same** direction for **1.76 km** at **60 km/h**.

Direction	Distance
Travel time	Speed

F Turn **130° right**.
Drive at **60 km/h** for **53 s**.

Direction	Distance
Travel time	Speed

4 ***Which activity are you most likely to do from your current location?***

a ***Cook a barbecue***

b ***Visit your old high school***

c ***Kick a football in the park***

d ***Make a telephone call***

G The traffic is heavy and you are caught behind a truck. Your speed is down to **30 km/h** until the truck turns right into Cotham Road.

Direction	Distance
Travel time	Speed

H Now you can speed up to **60 km/h**. Drive for **2 min 9 s** then stop.

Direction	Distance
Travel time	Speed

5 ***You're hungry – which is the closest place to buy food?***

a ***Red Rooster***

b ***KFC***

c ***McDonald's***

d ***Hungry Jack's***

I Continue for **11.6 s** at **60 km/h** and then turn **130° right**.

Direction	Distance
Travel time	Speed

J Proceed for **1.1 km** at **45 km/h** and then veer **40° left.**

Direction	Distance
Travel time	Speed

K Drive another **960 m** and you've reached your destination.

Direction	Distance
Travel time	Speed

6 ***In what field of study does this tertiary institution specialise?***

Destination data: ***Complete the table.***

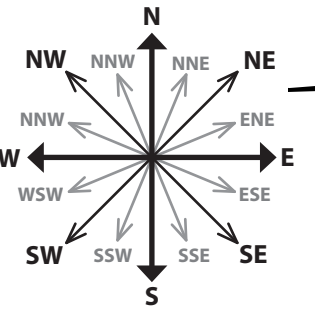
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

6 Sports and fast cars

Melways map 80



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- public telephone
- restaurant
- roundabout

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
Your starting position is 2.02 km in a direct line.
Clue 1: Measuring clockwise from due north, it is on an angle of 35° from the centre point.
Clue 2: It's where AFL football matches used to take place.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A Leave the ground and walk **260 m north** to the main road where your car is parked. (Don't include this distance in your total calculations.)

Direction	Distance
Travel time	Speed

B Drive for **2.7 km** at **80 km/h** in a **westerly** direction.

Direction	Distance
Travel time	Speed

C Turn **left** and proceed at **80 km/h** for **1 min 10 s.**

Direction	Distance
Travel time	Speed

D Turn **50° left** and proceed for **410 m** at **60 km/h**. Then **stop.**

Direction	Distance
Travel time	Speed

3 ***Which activity are you most likely to do from your current location?***

a ***Make a telephone call***

b ***Buy a fast food meal***

c ***Go shopping***

d ***Visit a caravan park***

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

6 Sports and fast cars (cont'd)

E Proceed for **925 m** at **70 km/h** and **stop**.

Direction	Distance
Travel time	Speed

4 ***Which activity are you most likely to do from your current location?***

a ***Visit a shooting range***

b ***Play a game of indoor tennis***

c ***Shop at a large hardware store***

d ***Visit the Sandown International Motor Raceway***

F Continue on the main road for **2.98 km** at **80 km/h.**

Direction	Distance
Travel time	Speed

G Turn **left**. Follow the road for **645 m.**

Direction	Distance
Travel time	Speed

H Veer **left** at the roundabout and proceed for **2 min 10 s** at **60 km/h**.

Direction	Distance
Travel time	Speed

5 ***At how many schools on Jackson's Road would you expect to see primary age students?***

I Wait **1 min 10 s** at the traffic lights for the lights to turn green and then turn **left**.

Direction	Distance
Travel time	Speed

J Follow this road for **3.1 km** at **80 km/h**. On your left you will pass the Springvale Botanical Cemetery.

Direction	Distance
Travel time	Speed

K Turn **140° left** onto a major carriageway and proceed for **1 min 7 sec** at **80 km/h**.

Direction	Distance
Travel time	Speed

L Turn **right** and proceed **1 250 m** at **20 km/h** to Gate 15.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

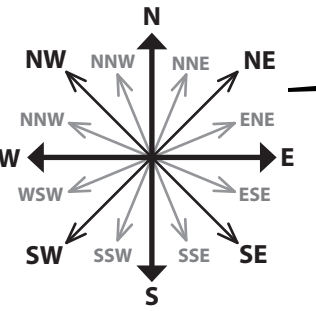
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

7 Sporting jaunt

Melways map 28



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- P carpark
- H hotel
- restaurant
- roundabout

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete:
Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
It's* 1.57 km *in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 216° from the centre point.

Clue 2: Practice makes perfect.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A Collect your car from the nearest roundabout. and drive in a **north-easterly** direction at **50 km/h** for **43.4 s**.

Direction	Distance
Travel time	Speed

B Turn **left** and keep driving at **50 km/h** for **260 m**.

Direction	Distance
Travel time	Speed

C Veer **44°** to the **right**. Unfortunately the traffic has become heavier and you can only average **40 km/h**. After **2 min 26 s**, you come to a T-intersection.

Direction	Distance
Travel time	Speed

D Turn **right** and drive **65 m**. Your speed is **20 km/h**.

Direction	Distance
Travel time	Speed

E Turn **left**. Drive at **50 km/h** for **500 m.**

Direction	Distance
Travel time	Speed

F Turn **right**. Continue for **516 m at 50 km/h** in an **easterly** direction through the roundabouts to the main road.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

7 Sporting jaunt (cont'd)

G Proceed at **60 km/h** to the intersection of Napier Crescent and Pascoe Vale Road (**Map 28 J2**) by the shortest available route. List the streets and distances you travel.

Direction	Distance
Travel time	Speed

H From this intersection, drive along Pascoe Vale Road in a **northerly** direction for **28.4 s** at **60 km/h** and **stop**.

Direction	Distance
Travel time	Speed

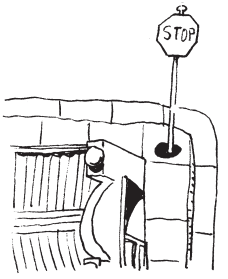
3 ***Which activity are you most likely to do from your current location?***

a ***Have lunch at the Curry Hut***

b ***Cook a barbecue***

c ***Visit the Lapidary Club***

d ***Have lunch at the Cross Keys Hotel***



I Take the shortest route to the Essendon Cricket Ground, travelling at **50 km/h**. List the streets and distances you travel. Park at the corner of Brewster and Napier Streets and take a tour through the Hall of Fame.

Direction	Distance
Travel time	Speed

4 ***What other sports facilities are featured in the immediate area?***

J From the car park, drive **south** at **50 km/h** for **490 m.**

Direction	Distance
Travel time	Speed

K At the roundabout, veer **right 36°** onto the main road. Continue at **60 km/h** for **1 min 29 s**.

Direction	Distance
Travel time	Speed

L Turn **left** and continue for **560 m** at **60 km/h**. You're right outside your final destination.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

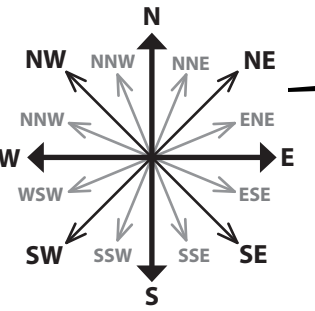
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

8 Medical mystery tour

Melways map 70



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- roundabout
- railway station
- P car park

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
It's* 100 m *in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 287° from the centre point.

Clue 2: They care about your health.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A You've been invited to give a presentation at a community health centre. To get to the health centre, start by driving **north** at **70 km/h** for **1 220 m** to the next major intersection.

Direction	Distance
Travel time	Speed

B Turn **right** and continue at **70 km/h** for **1.23 km**.

Direction	Distance
Travel time	Speed

C At the intersection, turn **left.** Keep driving at **70 km/h** and take the last right turn before the train station.

Direction	Distance
Travel time	Speed

D Reduce speed to **50 km/h**. Turn **right** after **260 m.** Stop at the community health centre which is **65 m** away on your **right**.

Direction	Distance
Travel time	Speed

E Now it's off to a 'Meals on Wheels' distribution centre to talk about nutrition. The centre is located at **Map 70 E0**. Take the shortest route to the centre. Your speed will be **50 km/h** on suburban roads and **70 km/h** on the two main roads. List the streets, distances, times and speeds you drive.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

8 Medical mystery tour (cont'd)

F There's a pizza shop at the intersection of Huntingdale and Grenfell Roads (**Map 70 A1**) but it's not shown on your map. Take the shortest route there. Your average speed will be **60 km/h**. List the streets and distances you drive and the time it takes. Watch out for that 'one-way' section of road!

Direction	Distance
Travel time	Speed

G After pizza, it's off to a secondary college to run a first aid course. Start by driving **1.4 km south** at **70 km/h** on the main road.

Direction	Distance
Travel time	Speed

H Turn **left** onto the freeway where the speed limit is **100 km/h**. Continue at **100 km/h** for **1 min 48 s.**

Direction	Distance
Travel time	Speed

I Take the **400 m** exit ramp at **80 km/h**.

Direction	Distance
Travel time	Speed

J Turn **right** and proceed at **70 km/h** for **1.82 km**.

Direction	Distance
Travel time	Speed

K Turn **left** and drive at **50 km/h** for **500 m**, then park.

Direction	Distance
Travel time	Speed

3 ***What is the name of the former school on your left?***

L Drive the most direct route to the 'Sir John Monash Private Hospital' (**Map 70 C12**), taking the final measurement to the corner of Clayton Rd and Dixon St. The speed limit is **70 km/h**. List the streets and distances you drive and the time it takes.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

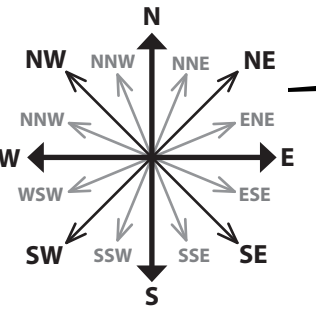
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

9 Skating round schools

Melways map 18



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- ☎ public telephone
- **P** car park
- roundabout

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
It's 2.62 km in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 28° from the centre point.

Clue 2: It has its ups and downs.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A The car is parked **60 m** due **west** in the adjacent street. Don't include this walk in your driving calculations.

Direction	Distance
Travel time	Speed

B Drive due **south** to the roundabout at **60 km/h**.

Direction	Distance
Travel time	Speed

C Veer **65° left**. Proceed for **600 m** at **60 km/h.**

Direction	Distance
Travel time	Speed

D Turn **right** at the roundabout. Continue at **60 km/h** for **1 min 24 s.**

Direction	Distance
Travel time	Speed

E Turn **70°** to the **left**. Drive at **60 km/h** for **800 m** and park.

Direction	Distance
Travel time	Speed

3 ***Describe the activities available at the place at which you have parked.***

F Continue driving in the **same** direction at **60 km/h** to the T-intersection at the end of the street.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

9 Skating round schools (cont'd)

G Turn **right**. Continue **south-east** at **60 km/h** to the roundabout.

Direction	Distance
Travel time	Speed

H Head almost due **west** from the roundabout. Proceed for **1 min 14 s** at **60 km/h**

Direction	Distance
Travel time	Speed

I Turn **left**. The public telephone on the corner is out of order. Travel **south** at **60 km/h** until you reach another public telephone. Park at the corner of Queen Street.

Direction	Distance
Travel time	Speed

J From here you have to visit two schools: Holy Name Catholic Primary School (Robb St) and Preston Primary School (Tyler St). The speed limit is **50 km/h** all the way. Follow the shortest, most convenient route to the two schools. List the streets and distances you drive and the time it takes.

Direction	Distance
Travel time	Speed

K After leaving the second school drive to the Preston Town Hall (**Map 18 G12**). The speed limit in High Street is **60 km/h**.

Direction	Distance
Travel time	Speed

L Drive **40 m** past the Town Hall to the intersection. Turn **right**.

Direction	Distance
Travel time	Speed

M Proceed at **60 km/h** for **1 min 15 s** and turn **right**. Continue for **2.3 km** and **stop**.

Direction	Distance
Travel time	Speed

N Now work out the location of your final destination. It is **10° east** of **due north**, **720 m** in a direct line from your current location. Drive the final **54 s** at **60 km/h** and **stop**.

Direction	Distance
Travel time	Speed

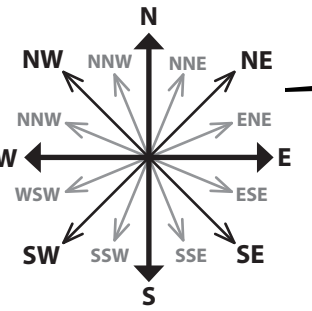
Destination data: ***Complete the table.***

Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

10 Off to the skate park Melways map 45



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- public telephone
- restaurant
- roundabout

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
It's 2.0 km in a direct line.
Clue 1: Measuring clockwise from due north, it is on an angle of 324° from the centre point.
Clue 2: It's new from old.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A Drive **194 m** in an **easterly** direction to the main road at **25 km/h.**

Direction	Distance
Travel time	Speed

B Turn **left.** Drive **194 m** at **70 km/h.**

Direction	Distance
Travel time	Speed

C Turn **right**. Follow this main road at **70 km/h** for **495 m**.

Direction	Distance
Travel time	Speed

3 ***Could you have reached this point more quickly by an alternative route? Explain.***

D Turn **left.** Drive for **40 s** at **60 km/h** and then accelerate to **70 km/h** for the next **1 min 39 s**.

Direction	Distance
Travel time	Speed

E At the intersection, turn **142° right** and proceed at **70 km/h** for **118 m** and park.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

10 Off to the skate park (cont'd)

4 ***Which activity are you most likely to do from your current location?***

a ***Have a snack at McDonalds***

b ***Use the public telephone***

c ***Visit the scout centre***

d ***Pay a telephone account***

F Get back in the car and proceed in a **south-westerly** direction for **3.35 km** at **70 km/h**.

Direction	Distance
Travel time	Speed

5 ***How many food outlets did you pass along this section? Name each one.***

G Veer **57° left** and proceed for **688 m** at **60 km/h**.

Direction	Distance
Travel time	Speed

6 ***At this point you are at the right angle of a triangle bounded by three main carriageways. What is the area in m^2 of this triangle?***
What is the angle at the intersection of High Street South and Barkers Road?

H Now work out how to get to the Hawthorn Football Club Social Club in Ruby Street (**Map 45 C10**) in the shortest possible time. You can drive at **60 km/h** on main roads and **50 km/h** on minor roads. List the streets, distances and speeds you drive and the time it takes.

Direction	Distance
Travel time	Speed

I It's off to the Skate and BMX Park, your final destination. But where is it?

Direction	Distance
Travel time	Speed

It's x kilometres in a direct line, and $y°$ east of due north. Solve these two equations for x and y.

$5x + 124 = 134.3$ $\qquad$ $20.6 + \frac{y}{2} = 82.6$

Work out how to get there in the shortest possible time. You can drive at **60 km/h** on main roads and **50 km/h** on minor roads. List the streets, distances and speeds you drive and the time it takes.

Destination data: ***Complete the table.***

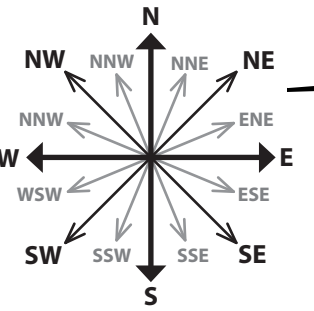
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

11 Dented enroute

Melways map 80



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- restaurant
- **P** car park
- roundabout

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
It's 2.46 km in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 349° from the centre point.

Clue 2: Do you get the point?

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A To reach your car, walk **650 m** along the path past the Senior Citizens and Community Centre car park. Begin your calculations from where your car is parked at the corner of Monash Drive and Florey Crescent.

Direction	Distance
Travel time	Speed

B Drive **130 m south** at **50 km/h** to the main road and turn **left**.

Direction	Distance
Travel time	Speed

C Proceed for **860 m** at **80 km/h** and enter the freeway. The on-ramp and off-ramp speed limit is **80 km/h**, but on the freeway the speed limit increases to **100 km/h**.

Direction	Distance
Travel time	Speed

D Exit the freeway at Jacksons Road, using the off-ramp.

Direction	Distance
Travel time	Speed

E Drive in a **southerly** direction at **70 km/h** for **409 m** and turn **right**.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9

11 Dented enroute (cont'd)

F It's time for lunch at Taco Bill's. Proceed for **2 min 11 s** at **80 km/h** and turn **left.**

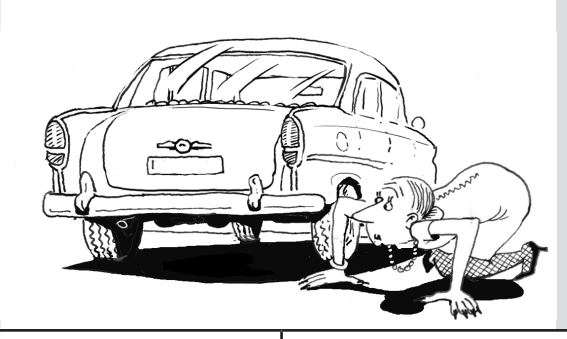
Direction	Distance
Travel time	Speed

G Use the service road to access the car park at Taco Bill's from Redpath Crescent. It's a **15 s** drive at **50 km/h**.

Direction	Distance
Travel time	Speed

H When you return to your car after lunch, you discover that it has been dented at the back of the driver's side. Continue your journey from the intersection of Redpath Crescent and the Princes Highway. Drive at **80 km/h** for **2 min 4 s** in a **south-easterly** direction.

Direction	Distance
Travel time	Speed



I Turn **left** and reduce your speed to **60 km/h**. Park your vehicle after travelling **129 m**. Your destination is the main office of the building on the west side of the road.

Direction	Distance
Travel time	Speed

3 ***What is the most likely reason for this stop?***

J Complete a U-turn and drive **129 m** to the main road.

Direction	Distance
Travel time	Speed

K Turn **right**. Drive for **40.6 s** (you already know the speed limit).

Direction	Distance
Travel time	Speed

L Turn **left.** Drive **for 1 min 48 s** at **60 km/h**. Turn **right**. Drive for **750 m.**

Direction	Distance
Travel time	Speed

M Turn **left** and then almost immediately **right**. Drive for **120 m** at **70 km/h** and you've reached your destination.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

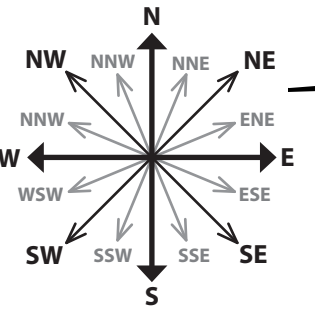
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

12 Sydney sojourn

Sydways map 64



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- roundabout
- railway station
- restaurant

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
It's* 1.13 km *in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 62° from the centre point.

Clue 2: They set the rates.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A Drive **south-west** for **30 m** at **60 km/h** and turn **right**.

Direction	Distance
Travel time	Speed

B Drive **140 m** to the main road and then turn **left.**

Direction	Distance
Travel time	Speed

C Continue at **70 km/h** for **1 min** and turn **145° left.**

Direction	Distance
Travel time	Speed

D After **80 m** at **60 km/h**, turn **right**.

Direction	Distance
Travel time	Speed

E Proceed for **1 min 9 s** at **60 km/h**.

Direction	Distance
Travel time	Speed

F Turn **right** – you're lucky, it's a one-way street and you're going the right way. Proceed to the next intersection at **40 km/h**.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

12 Sydney sojourn (cont'd)

G Your next stop is the Stanmore Library located near the Stanmore Railway Station. The entry is in Douglas Street and it's **50 km/h** all the way. List the streets and distances you drive and the time it takes.

Direction	Distance
Travel time	Speed

H After returning an overdue book, drive **south-west** at **60 km/h** to the corner and turn **132° right**.

Direction	Distance
Travel time	Speed

I Drive **60 m** and take a **44°** turn to the **right**. Proceed to the first roundabout and turn **left**.

Direction	Distance
Travel time	Speed

J Now it's back to **50 km/h** until you reach the main road **422 m** to the **north**. Turn right and drive **380 m** at **70 km/h**.

Direction	Distance
Travel time	Speed

3 ***You've reached a well-known fast food establishment. What is it?***

K Continue **east** on this main road for **8.35 s** at **70 km/h**. Then veer **35°** to the **left**.

Direction	Distance
Travel time	Speed

L Drive at **70 km/h** along this main road for **50 s**. Turn **left**.

Direction	Distance
Travel time	Speed

M Proceed at **60 km/h** for **97 m** and veer **left**. Continue for a further **793 m**.

Direction	Distance
Travel time	Speed

4 ***What sporting venue did you pass? What type of events does it host?***

N Take a sharp **left** turn and proceed for **41 s**. Turn **right**. Continue for **280 m.**

Direction	Distance
Travel time	Speed

O Turn **left** at the intersection and drive **160 m**. The building you're after is on your left.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

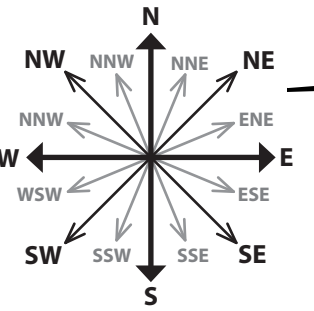
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

13 From ABC to TV

Sydways map 35



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- restaurant
- P car park
- roundabout

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
It's* 100 m *in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 266° from the centre point.

Clue 2: An alphabet mystery.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A Drive **north-west** for **60 m** at **60 km/h**, then turn **right**. At the end of the street, turn **right**.

Direction	Distance
Travel time	Speed

B Drive for **9 s** and then turn **left**. **20 s** later, turn **right**. Turn **left** **12 s** after that. It's been a zig-zagging route so far.

Direction	Distance
Travel time	Speed

C Drive **270 m** over the bridge and turn **right** into a short street leading to a sporting venue. The speed limit has dropped to **50 km/h** in this street. Have a break and play a game or two.

Direction	Distance
Travel time	Speed

3 ***What did the bridge cross?***

4 ***What sport did you play?***

D Turn **left** into Talus Street and go back up to Dalleys Road. It's still **50 km/h** until you reach the T-intersection, where it reverts to **60 km/h**. Turn **right**. Almost immediately, turn **left**.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

13 From ABC to TV (cont'd)

E Continue at **60 km/h** for **180 m.** Turn **right** into a street with a **50 km/h** limit.

Direction	Distance
Travel time	Speed

F After **100 m**, turn **left** and continue at **40 km/h** to the end of the entry road to the Centre.

Direction	Distance
Travel time	Speed

5 ***What is a likely reason for this stop?***

G Return via the entry road, turn **left** to proceed at **50 km/h** in a **southerly** direction to the T-intersection at the end of the street. Turn **right**.

Direction	Distance
Travel time	Speed

H Proceed at **60 km/h** to the second roundabout. Turn **left** and drive **210 m**. Then veer **50° left** at the Y-intersection.

Direction	Distance
Travel time	Speed

I Proceed along this street in a **south-easterly** direction for **963 m.** Turn **left**. After **110 m**, turn **left** again. Continue on the highway for **196 m** at **80 km/h**.

Direction	Distance
Travel time	Speed

J Take the road leading slightly **east** of **north** (it's a **42°** change of direction). Proceed on this road at **60 km/h** until you reach a T-intersection. Turn **left**.

Direction	Distance
Travel time	Speed

K Drive for **10.6 s** at **70 km/h**. Turn **right** at the intersection onto a main road. Drive for **1.31 km**. When the road reaches the freeway overpass, the speed limit changes from **70 km/h** to **80 km/h**.

Direction	Distance
Travel time	Speed

Direction	Distance
Travel time	Speed

L Turn **left** and proceed for **240 m** at **60 km/h**. Park your car.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

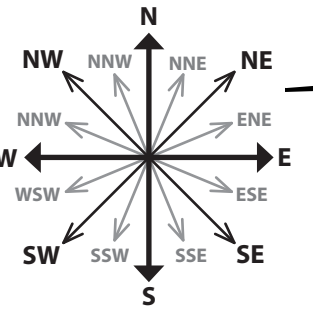
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

14 Traffic light trek

Melways map 28



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

- **P** car park
- roundabout
- church

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
It's* 1.86 km *in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 165° from the centre point.

Clue 2: The highlight is the 'Grand Parade'.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A Drive **north-west** for **1.44 km** at **60 km/h**. At the main road intersection, veer **44°** to the **right**.

Direction	Distance
Travel time	Speed

B Proceed through the dog-leg for **880 m** and turn **left**.

Direction	Distance
Travel time	Speed

C Drive at **40 km/h** for **237 m** and park your vehicle.

Direction	Distance
Travel time	Speed

3 ***Which sport are you most likely to participate in at this stop?***

a ***Golf*** **b** ***Squash*** **c** ***Athletics*** **d** ***Tennis***

D Return to the main road and **stop**. Your next destination is **2.4 km** in a direct line **73° east** of due **north** from your current location. *Hint:* It's a school from which drivers could benefit. Follow the shortest route to your destination, at **60 km/h** all the way. Don't forget to keep track of the distance you drive.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

14 Traffic light trek (cont'd)

E Now proceed **west** for **1.2 km**. Turn **left** onto a main carriageway.

Direction	Distance
Travel time	Speed

F Continue for **1 min 56 s** at **60 km/h**. Then turn **right** at the intersection.

Direction	Distance
Travel time	Speed

4 ***How many different churches did you pass in the last 6 kilometres?***

G Drive **west** for **2 min 46 s** at **60 km/h** and turn **left**.

Direction	Distance
Travel time	Speed

5 ***You crossed over a river. What is its name and how wide is it at your point of crossing?***

H Continue for almost **1 km** to the roundabout and turn **left** into the park.

Direction	Distance
Travel time	Speed

I Follow the road for **450 m** at **20 km/h** to the car park adjacent to the visitors' centre. Park your car.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

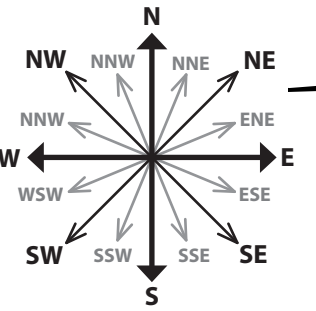
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

15 City circuit

Sydways map 64



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

P car park

roundabout

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
It's 960 m in a direct line

Clue 1: Measuring clockwise from due north, it is on an angle of 120° from the centre point.

Clue 2: You have to have heart.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

Use the tables to help organise your calculations.

A Proceed to the nearest main intersection and drive in a **westerly** direction for **1.16 km** at **60 km/h**.

Direction	Distance
Travel time	Speed

B Turn **right.** Drive to the end of this road and turn **left** at the T-intersection.

Direction	Distance
Travel time	Speed

C The speed limit on this carriageway and the one you will enter next is **80 km/h**. Drive for **16.1 s** and turn **right**.

Direction	Distance
Travel time	Speed

3 ***Measure the angle formed by the carriageway you were just on, and the one you are now entering. Regard the road you are entering as the hypotenuse of a right-angled triangle, and the distance you drove on Parramatta Road as the length of one side.***

a ***What distance would complete the hypotenuse of this triangle?***

b ***What area of land does this triangle encompass?***

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

15 City circuit (cont'd)

D Using the distance you calculated for the hypotenuse as a guide, drive to the intersection nearest that distance. Turn **left**. Record the actual distance to the intersection, **not** the hypotenuse length you calculated.

Direction	Distance
Travel time	Speed

E The speed limit is back to **60 km/h**. Proceed for **8 s** and turn **right**, then **20.8 s** and turn **left**.

Direction	Distance
Travel time	Speed

F Continue for **142 m** and turn **right**. Drive **142 m** and park.

Direction	Distance
Travel time	Speed

4 ***Where are you? Suggest why you have stopped.***

G Drive **142 m** to the **northerly** end of the street and stop to consult your map. Your next destination is 'Wattle House – Accommodation.' (**Map 64 R2**) Follow the shortest route there. Assume **60 km/h** all the way. *Hint:* Watch the road signs and thoroughfare access.

Direction	Distance
Travel time	Speed

5 ***How long did it take you to get from the T-intersection at the corner of Hutchinson Lane and Hutchinson Street to Wattle House?***

H From Wattle House, proceed at **60 km/h** along Hereford St and turn **left** into Minogue Cres.

Direction	Distance
Travel time	Speed

I Continue **south-east** at **70 km/h** for **28.4 s**. Turn **right**. It's **80 km/h** on this main road. Drive **510 m** and turn **left**.

Direction	Distance
Travel time	Speed

J After **650 m** at **60 km/h** you've reached your destination. It's on your **right**.

Direction	Distance
Travel time	Speed

Destination data: ***Complete the table.***

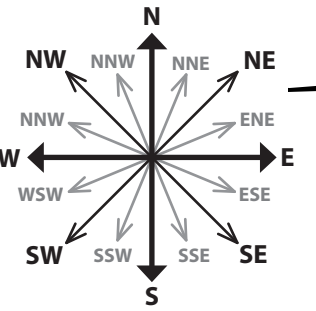
Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

16 Racing round about

Melways map 80



Your task is to follow these directions on the given map to reach a final destination. You will need to solve problems about speed, distance and time and complete scale calculations to work out your position on the map.

Keep track of each distance you cover, and your speed for each distance – at the end, you'll be asked to calculate your total distance, total driving time and average speed.

Map symbols you need:

P car park

roundabout

1 ***Look at the scale on your map.***
Measure the divisions and reproduce them below.

Now label the scale above with the distance in real life indicated by each division. Include the units.

Complete: Distance in real life in metres ÷ ____ = length in centimetres on the map

2 ***Begin at the map's centre point.***
It's 2.47 km in a direct line.

Clue 1: Measuring clockwise from due north, it is on an angle of 148° from the centre point.

Clue 2: It's all about water.

Tip: *Use your common sense in your calculations – for example, is it practical to be measuring to thousandths of a centimetre on your map?*

3 ***What game could you be there to play?***

Use the tables to help organise your calculations.

A Exit to the main road and begin your measurements from here. Drive **north** at **70 km/h** for **300 m**.

Direction	Distance
Travel time	Speed

B Veer **left** through the roundabout. Proceed for **1 min 56 s** at **70 km/h**.

Direction	Distance
Travel time	Speed

C Veer **left** onto a major carriageway. The speed limit for the **500 m** on-ramp is **80 km/h**. For the next **1.36 km**, the speed limit is **100 km/h**.

Direction	Distance
Travel time	Speed

D Take the off-ramp at **80 km/h**. Veer **left** and continue at **80 km/h** in a **westerly** direction for **1.16 km**. Turn **left**.

Direction	Distance
Travel time	Speed

E Find your way to the Mulgrave Art Gallery (**Map 80 B2**). Once you leave the main road, the speed limit is **60 km/h**. List the streets, distances and speeds you drive and the time it takes.

Direction	Distance
Travel time	Speed

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

16 Racing round about (cont'd)

F Now get yourself to the entry of the Waverley Gardens shopping centre (**Map 80 J5**) in the shortest time you can.

Main roads in this area have **80 km/h** speed limits, freeways have **100 km/h** limits with **80 km/h** access ramps, and all other roads have **60 km/h** speed limits.

List the streets, distances and speeds you drive and the time it takes.

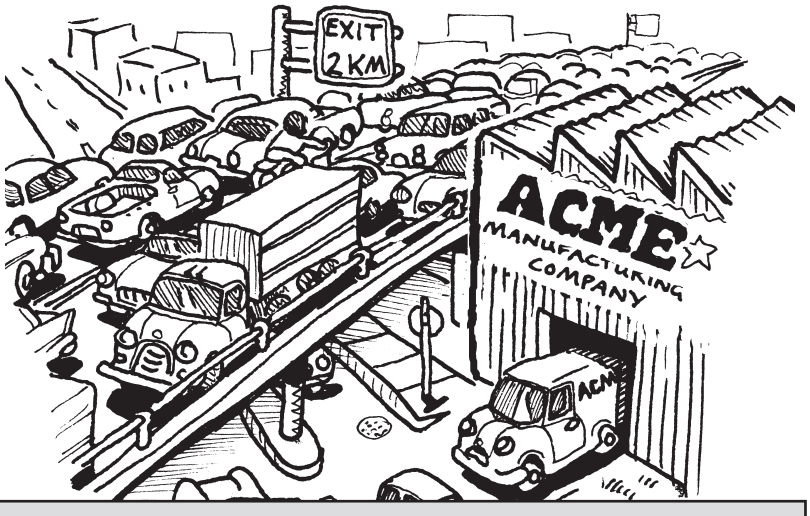
Direction	Distance
Travel time	Speed

G Now proceed **due south**, crossing the main road from the shopping centre entry, for **2.06 km** at **60 km/h**. Then **stop**.

Direction	Distance
Travel time	Speed

4 ***What is the most likely reason for this stop?***

- **a** ***To have lunch at a fast food outlet***
- **b** ***To pay an auto club membership fee***
- **c** ***To visit a relative at a nursing home***



H Proceed to the main road. Turn **right** and drive in a **north-west** direction at **80 km/h** for **40.6 s.**

Direction	Distance
Travel time	Speed

I Turn **left.** Continue on this road at **60 km/h** for **1 min 5.8 s**.

Direction	Distance
Travel time	Speed

J Turn **right**. Drive for **1.12 km** at **50 km/h** and park. You have reached your destination.

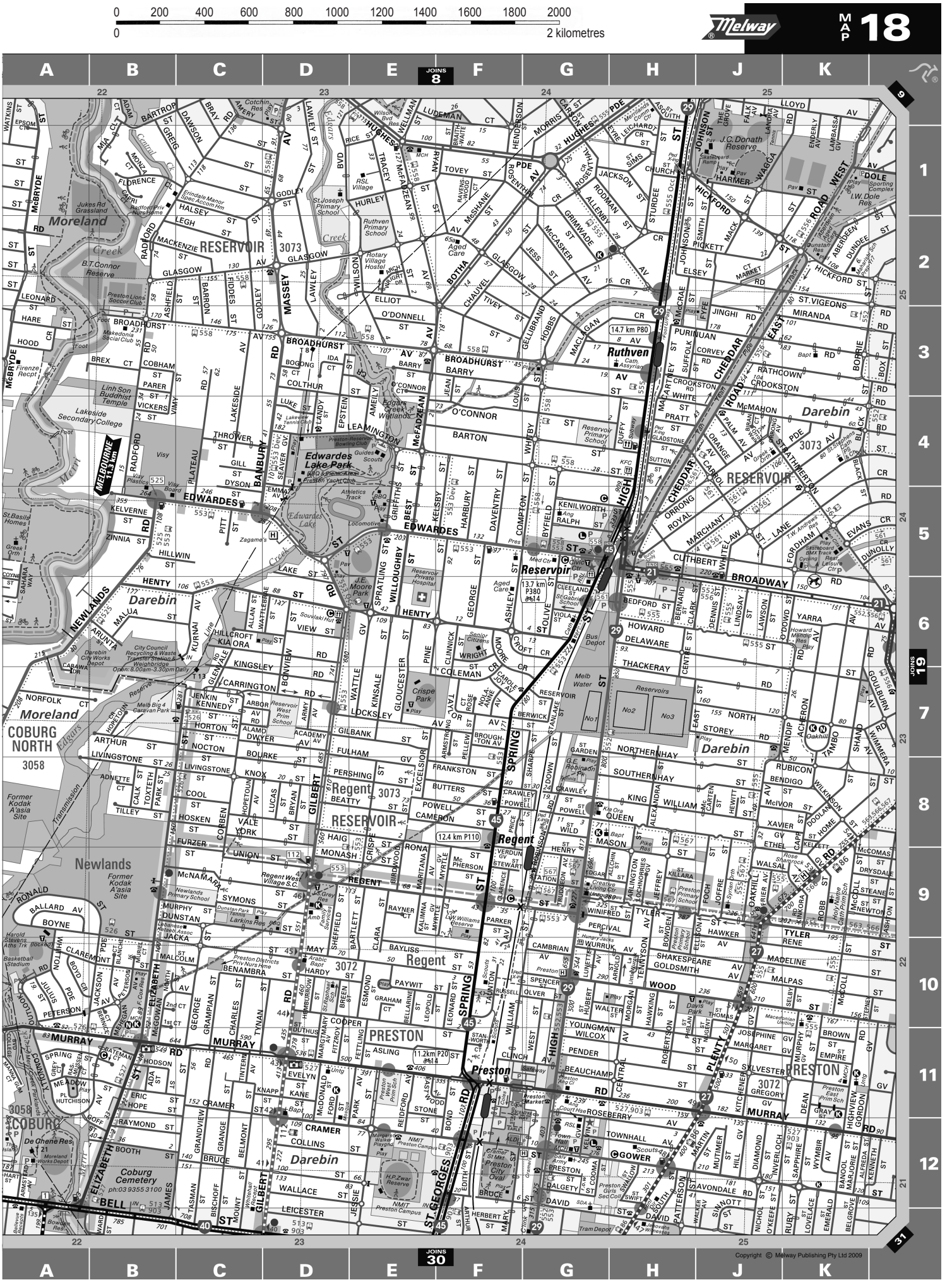
Direction	Distance
Travel time	Speed

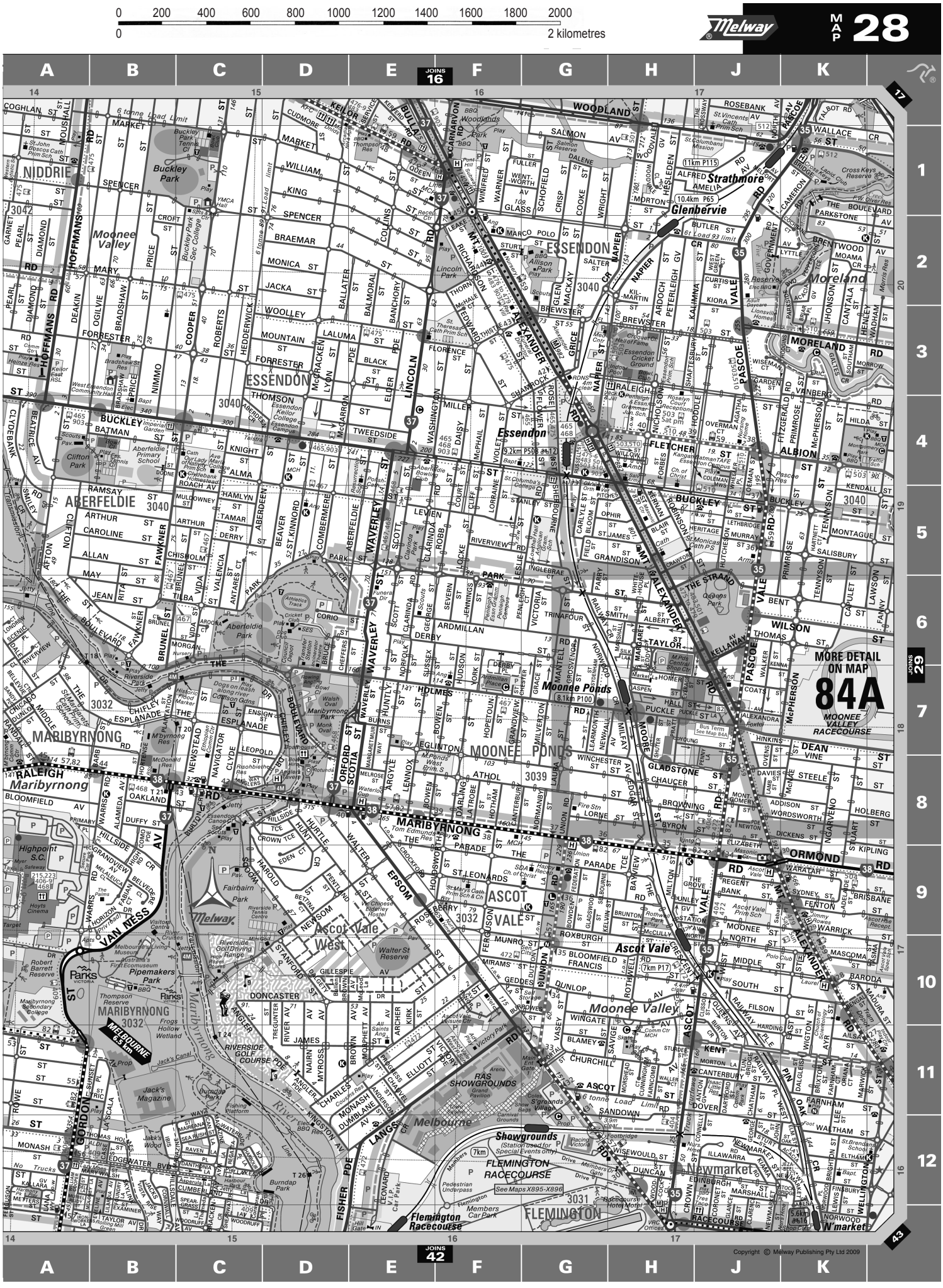
Destination data: ***Complete the table.***

Name of destination and map coordinates (map no., grid reference)	Total distance driven (km)	Total driving time (min, s)	Average driving speed (km/h)

© G.A.T.E.WAYS Publications and Blake Education 2010

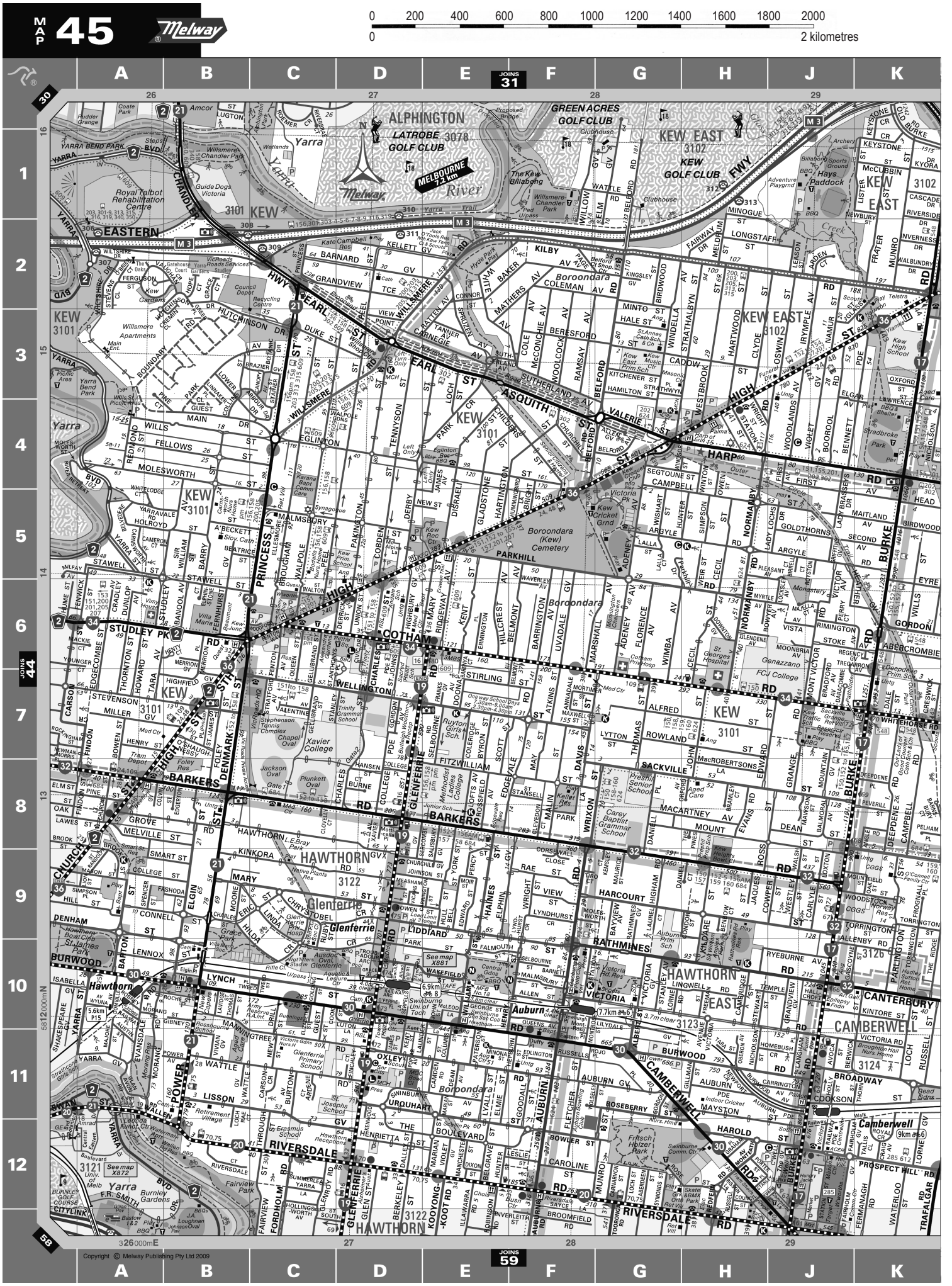
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

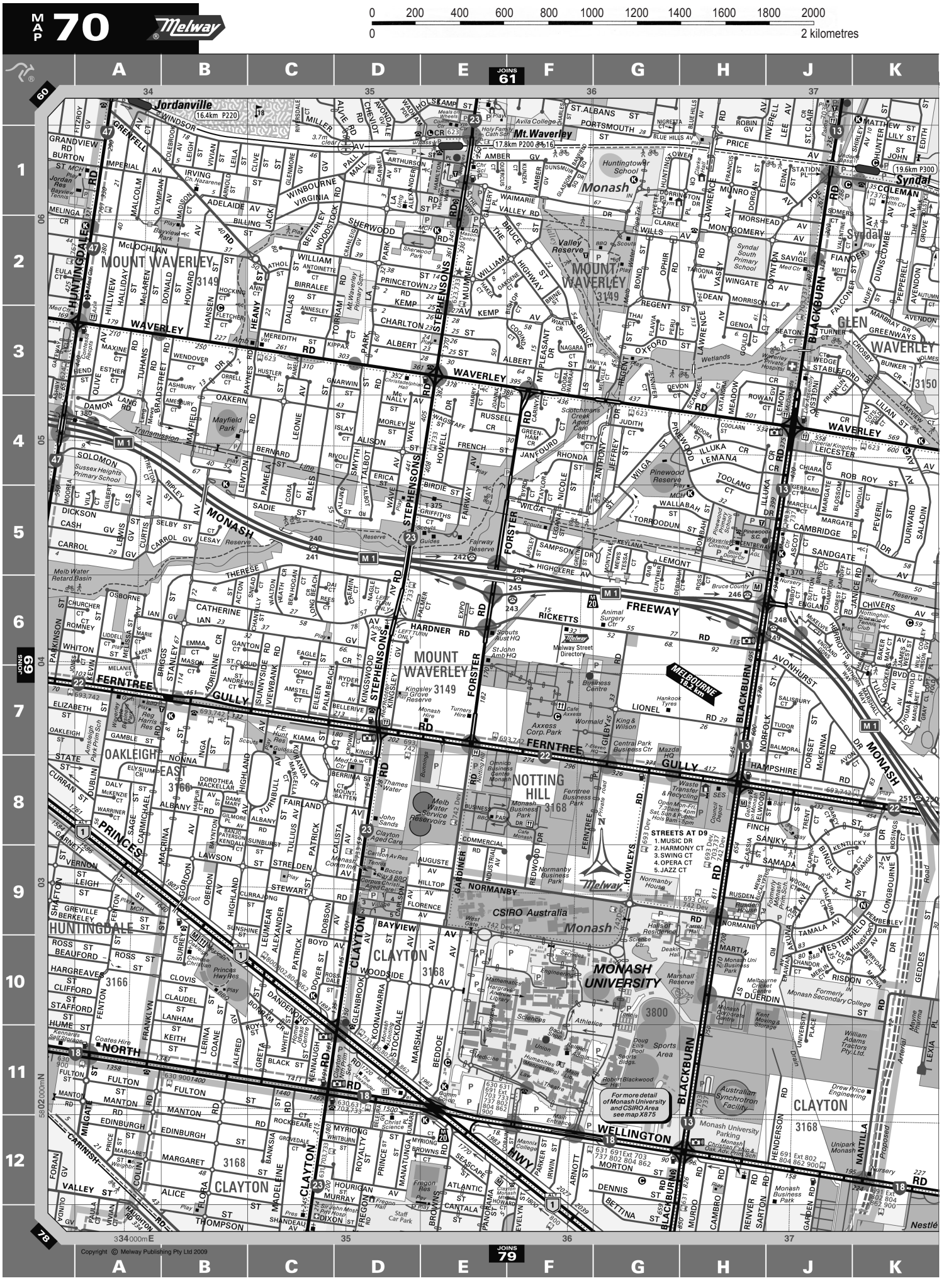

0 200 400 600 800 1000 1200 1400 1600 1800 2000
0 2 kilometres
Melway MAP 18
A B C D E F G H J K
JOINS 8
JOINS 19
JOINS 30
22 23 24 25
1 2 3 4 5 6 7 8 9 10 11 12
Moreland
RESERVOIR 3073
Darebin
Edwardes Lake Park
Ruthven
Darebin
RESERVOIR 3073
COBURG NORTH 3058
Moreland
Newlands
Regent 3073
RESERVOIR
Regent
Regent 3072
PRESTON
Preston
PRESTON 3072
COBURG 3058
Darebin
Coburg Cemetery ph:03 9355 3100
Darebin
Preston City Oval
Darebin
Reservoir
Darebin
Darebin
Darebin



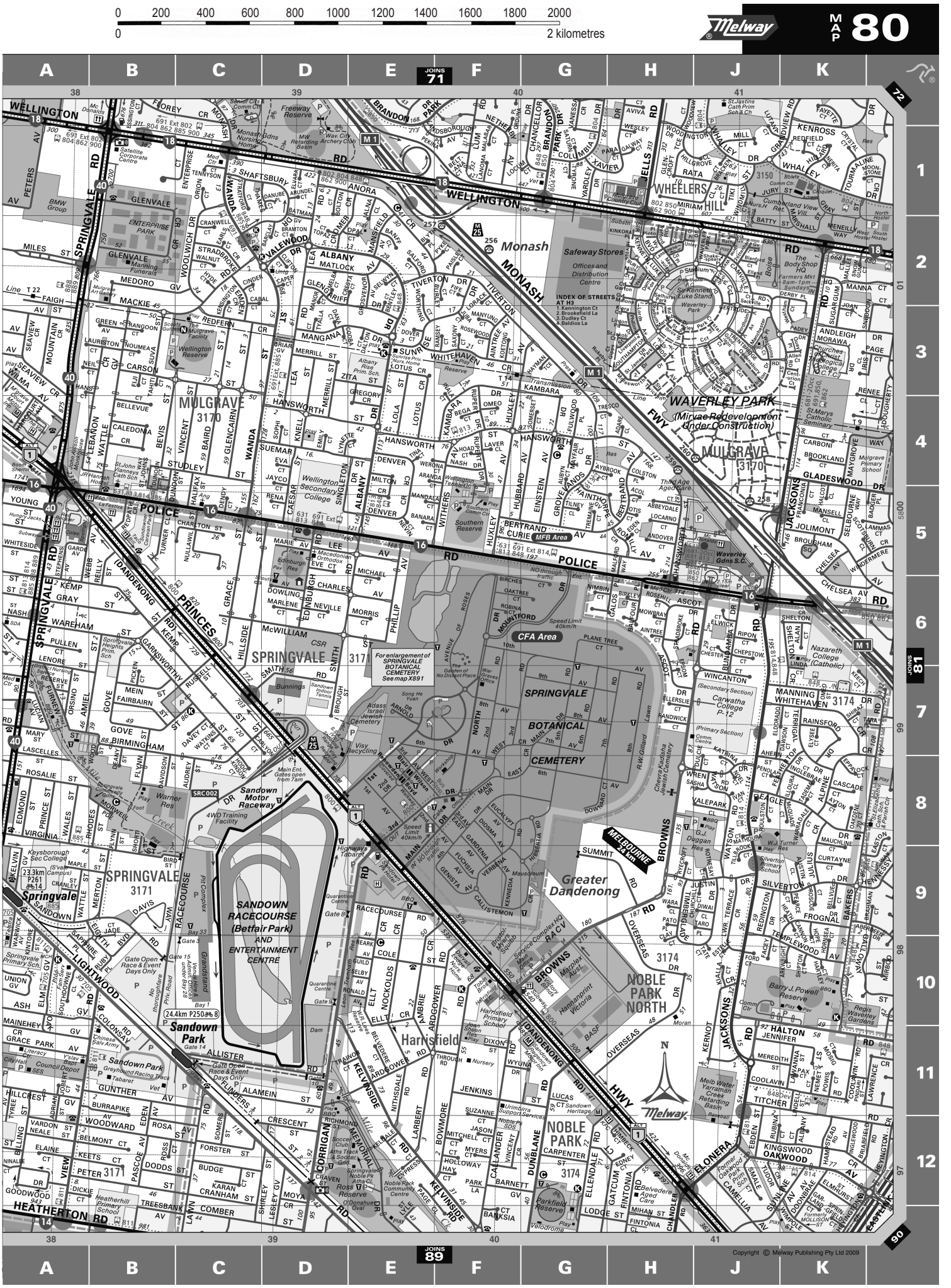
Map copyright © Melway Publishing Pty Ltd 2009

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys


MAP 45
Melway
0 200 400 600 800 1000 1200 1400 1600 1800 2000
0 2 kilometres
A B C D E JOINS 31 F G H J K
ALPHINGTON
LATROBE 3078
GOLF CLUB
GREEN ACRES
GOLF CLUB
KEW EAST
3102
KEW
GOLF CLUB
MELBOURNE
7.2 km
Yarra
River
Melway
3101 KEW
EASTERN
HWY
CHANDLER
EARL ST
WILLSMERE
ASQUITH ST
HIGH ST
PRINCESS ST
COTHAM
HAWTHORN
Glenferrie
RIVERSDALE
Boroondara
(Kew)
Cemetery
PARKHILL
Auburn
CAMBERWELL
Camberwell
HARP
BARKERS
WELLINGTON
STUDLEY PK
RD
Yarra
Yarra Bend Park
Royal Talbot
Rehabilitation
Centre
Guide Dogs
Victoria
Willsmere
Chandler Park
Kew
Gardens
Kew
Cemetery
Victoria Pk
Kew Cricket
Grnd
Kew Rec
Res
Kew Prim.
School
Xavier
College
Chapel
Oval
Jackson
Oval
Plunkett
Oval
Gate 1
Stephenson
Tennis
Complex
Trinity
Grammar
School
Ruyton
Girls Sch.
Methodist
Ladies
College
Carey
Baptist
Grammar
School
Presh.
Senior
School
Aged
Care
St.
Georges
Hospital
Genazzano
FCJ College
Kew
High
School
Kew
Primary
School
Kew East
Prim Sch
St Anne's
Parish Sch
Glen Iris
Primary
School
Josepha
School
Erasmus
School
Hawthorn
Grammar
School
Swinburne
Univ. of
Technology
Swinburne
Comm. Ctr.
Fritsch
Holzer
Park
Indoor Cricket
Broughton Hall
Nurs. Home
Kew
Cemetery
Yarra
F.R. SMITH
Burnley
Gardens
Fairview
Park
Univ.
of
Melb
Retirement
Village
Aged Care
Nurs. Hme
Nurs. Home
Boulevard
Shopping
Centre
Hawthorn
Arts Centre
Swinburne
Univ. of
Technology
Hawthorn
Aquatic &
Leisure Ctr
Glenferrie
Oval
Gardiners
Creek
Trail
Yarra
Trail
Creek
Billabong
Yarra
Boulevard
Yarra
Bend
Park
Picnic
Area
Main
Ent.
Willsmere
Apartments
Retirement
Village
Aged Care
Nurs. Hme
Nurs. Home
Kew
Cemetery
Kew
Cricket
Grnd
Kew
Rec
Res
Kew
Prim.
School
Kew
High
School
Kew East
Prim Sch
Kew
Primary
School
Kew
Cemetery

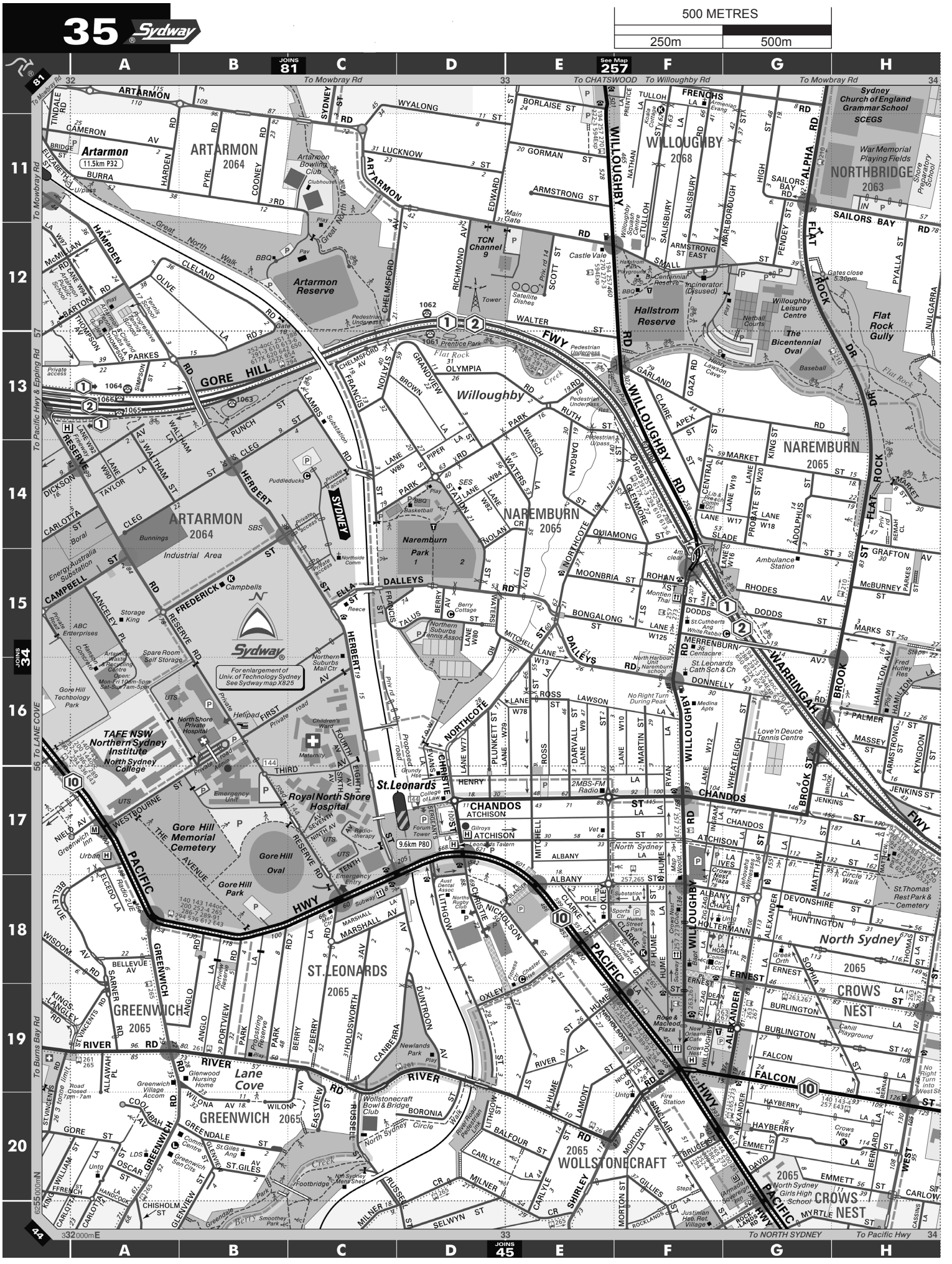


Map copyright © Melway Publishing Pty Ltd 2009



Map copyright © Melway Publishing Pty Ltd 2009

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys


35 Sydway
500 METRES
250m
500m
Artarmon
ARTARMON 2064
WILLOUGHBY 2068
NORTHBRIDGE 2063
NAREMBURN 2065
ARTARMON 2064
Willoughby
GREENWICH 2065
ST LEONARDS 2065
GREENWICH 2065
WOLLSTONECRAFT 2065
CROWS NEST 2065
North Sydney
Artarmon Reserve
Hallstrom Reserve
Flat Rock Gully
The Bicentennial Oval
Willoughby Leisure Centre
Naremburn Park
Gore Hill Oval
Gore Hill Park
Gore Hill Memorial Cemetery
Royal North Shore Hospital
TAFE NSW Northern Sydney Institute North Sydney College
St.Leonards
Greenwich Village
Greenwich Village Accom
Glenwood Nursing Home
Wollstonecraft Bowl & Bridge Club
Artarmon Bowling Club
Sydney Church of England Grammar School SCEGS
War Memorial Playing Fields
Bi-Centennial Reserve
Incinerator (Disused)
TCN Channel 9
Satellite Dishes
Tower
Willoughby Leisure Centre
Netball Courts
Baseball
Lawson Cove
Flat Rock Creek
Ambulance Station
Fire Station
Police Station
Sydney
Campbells
Bunnings
Industrial Area
Energy Australia Substation
Puddleducks
Northside Comm
Northern Suburbs Tennis Assoc
Berry Cottage
Storage King
ABC Enterprises
Spare Room Self Storage
Harsen Concrete
Artarmon Waste Recycling Centre
Open: Mon-Fri 10am-5pm Sat-Sun 7am-5pm
For enlargement of Univ. of Technology Sydney See Sydway map X825
Gore Hill Technology Park
North Shore Private Hospital
Helipad
Children's Ward
Maternity
Emergency Unit
Radio-therapy
Forum Tower
Leonards Tavern
Aust Dental Assoc
North Rugby Club
Sports Ctr
Chester Hse
Fitness First
Rose & Macleod Plaza
Crows Nest Plaza
Woolworths Metro
Medina Apts
Love'n Deuce Tennis Centre
Fred Hutley Res
St.Thomas' Rest Park & Cemetery
Cahill Playground
Newlands Park
Wollstonecraft Bowl & Bridge Club
Footbridge
Steps
Circle Walk
Bike Path
Pedestrian Underpass
Pedestrian Link
Private Road
Private access
Private road
Proposed road
Priv rd
Priv cl
Subway
Gates close 5.30pm
No Right Turn During Peak
4m clear
9.6km P80
11.5km P32
Artarmon
Great North Walk
Main Gate
Clubhouse
Play
Pav
BBQ
Tennis
Basketball
Lib & Sch Comm Ctr
Lib & Sch Comm Ctr
Lib & Sch Comm Ctr

500 METRES
250m
500m
Sydway 64
J K L M N P Q R
JOINS 54
JOINS 74
30 To City West Link Rd
31 To City West Link Rd
To Glebe Point
To Bridge Rd
To PYRMONT
To ULTIMO & SYDNEY
To CHIPPENDALE & SYDNEY
To REDFERN
To City Rd
To Charles St
To Wilson St
To ENMORE
To Stanmore Rd
To ENMORE
To ST. PETERS & ARNCLIFFE
LILYFIELD 2040
ANNANDALE 2038
Leichhardt
LEICHHARDT 2040
ANNANDALE 2038
GLEBE POINT 2037
GLEBE 2037
FOREST LODGE 2037
Sydney
Harold Park Paceway
Jubilee Park
St Scholastica's College
Cohen Park
Annandale North Public School
Annandale Public School
Transfer Station (Weekend Tip)
Annandale Village
Annandale Community Centre
Annandale Nursing Home
Brendans Catholic School
Buddhist Monastery
Evan Jones Playground
Hunter Baillie Park (1888)
Old Annandale Town Hall
St Aidans Ang
North Annandale
Whites Creek Valley Park
Whites Creek Wetlands
Hinsby Park
Hogan Pk
Arthur (Paddy) Gray Reserve
Wattle House Accom
St James Prim School
St Helens Community Centre
Forest Lodge Public School
Glebe Public School
Zodiac Lodge
Sydney Grammar School
Orphan School Creek
Rec Res
Police & Comm Yth Cl
Sydney Water
Stormwater Drain
Stormwater Channel
Sydney 2006
University
St Johns Oval
St Johns College
Sancta Sophia College
Bridge Rd School
Camperdown Bowling Club
Camperdown Park
Camperdown Memorial Rest Park
Royal Prince Alfred Hospital
King George V Hospital
St Andrews College
St Andrews Oval
Queen Elizabeth II Rehab Centre
Queen Mary Bldg
Tennis Centre
Staff Parking
Staff Access Only
Bridge Rd Pedestrian Bridge
Marrickville
Stanmore
Stanmore Plaza
Stanmore Public School
Stanmore Nursing Home
Newington College
Newington Prep Sch
Johnson Oval
Old Boys Oval
Eve Sharpe Res
Montague Gardens
Weekley Park
Scouts & Kinder
St Michaels Catholic Prim Sch
Holy Family Convent
Crest Nursing Home
Cohen Park Tennis Courts
Whites Creek Valley Park
Playground
Whites Creek Wetlands
Sydney Water
Stormwater Drain
Stormwater Channel
Rec Res
Police & Comm Yth Cl

Selected answers

1 Mapping toolbox

2 e.g. Glasgow Avenue, Maclagan Crescent, High Street, Macartney Street

3 **d** Tovey

4 F2, G2, G1

5 The relationship between length on the map and distance in real life

6 500 m

7 1 900 m

8 **c** 980 m **d** Answers will vary according to route chosen. Via Elliot and McFadzean, approximately 1200 m

10 160 s (2 min 40 s)

11 400 m

2 Sink or swim!

1–3 Use the map to calculate – will depend on reproduction size of map.

4 **a** H3 **b** G6 **c** G9

5 **Starting point:** Map 18 B9 in the car park of the former Kodak A'asia Site

A Drive to the intersection of Elizabeth Street and Murray Road.

B Drive along Murray Road to the intersection with High Street.

C Drive along High Street to the intersection with Cheddar Road West.

D Drive along Cheddar Road West to the intersection with Hickford Street.

E Drive along Hickford Street to the intersection with Johnson Street.

F Turn left into Johnson Street, left into Elsey Street, right into McCrae Street and right into Purinuan Road. The public telephone is a short distance along Purinuan Road.

G Drive along Macartney Street to the intersection with High Street.

H Turn right into Edwardes Street and drive to the roundabout at the south end of Edwardes Lake.

I Proceed via Edwardes Street and Banbury Roads to the Emma Avenue car park adjacent to the Preston Yacht club.

Destination data

- Preston Yacht Club, Map 18 D4

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 11.1 km
- **Total driving time:** 13 min 36 s
- **Average speed:** 48.97 km/h

3 Horses and courses

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 28 K7, Moonee Valley Racecourse at the corner of the T-intersection of Dean and Stuart Streets.

A Drive along Stuart Street to the intersection with Ormond Road.

B Drive along Ormond Road to Mt Alexander Road.

3 Mt Alexander Road

C Drive along Ormond Road (becomes Maribyrnong Road) to the intersection with Van Ness Avenue.

D Drive along Van Ness Avenue to the roundabout.

E Drive along Warrs Road to the entry to Highpoint Shopping Centre.

F Drive into the car park.

4 **a** Shopping

G Drive along Warrs Road to the roundabout at Van Ness Avenue.

H Drive along Van Ness Avenue to the intersection with Maribyrnong Road.

5 The Maribyrnong River

6 Essendon Canoe Club, Sea Scouts

I Drive along Maribyrnong Road to the intersection with Epsom Road.

J Drive along Epsom Road to the Racecourse Hotel/Motel.

Destination data

- Racecourse Hotel/Motel, Map 28 H12

Note that answers will be affected by measurement precision and the rounding of calculations.

- Total distance driven: 9.34 km
- Total driving time: 10 min 28 s
- Average speed: 53.56 km/h

4 Monash mayhem

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 70 F12 at the main entry to Monash University

A Drive along Wellington Road to the intersection with Princes Highway.

3 They both bear the names of famous soldiers. Sir John Monash was an Australian military commander in World War 1 and the Duke of Wellington was a British general who led the army that defeated Napoleon at the Battle of Waterloo.

B Drive along Princes Highway to the intersection with Clayton Road.

C Drive along Clayton Road (becomes Stephensons Road) to the intersection with Waverley Road.

4 Two – D11, D7

D Drive along Waverley Road to the intersection with Huntingdale Road.

5 Azia

E Turn left into Huntingdale Road.

F Drive to the on-ramp to Monash Freeway.

G Drive along Monash Freeway to the off-ramp for Blackburn Road.

H Drive along the off-ramp to Blackburn Road.

I Drive along Blackburn Road to the entry to the car park of the Australian Synchrotron Facility.

6 Marshall Reserve

J Drive into the car park.

Destination data

- Australian Synchrotron Facility, Map 70 H11

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 11.92 km
- **Total driving time:** 9 min 57 s
- **Average speed:** 71.83 km/h

5 Cruising Boroondara

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 45 B1 at the Guide Dogs Victoria centre

A Drive along Chandler Highway to the intersection with Princess Street and Earl Street.

B Drive along Princess Street to the intersection with High Street.

C Drive along High Street to the intersection with Belford Road.

3 **b** Watch a game of cricket.

D Continue along High Street to the intersection with Burke Road.

E Drive along Burke Road to the public telephone just past the Harp Road intersection.

4 **d** Make a telephone call.

F Continue along Burke Road to the Cotham Road intersection.

G Continue along Burke Road to the Hungry Jack's in the block past the Prospect Hill Road intersection.

5 **d** Hungry Jack's

H Continue along Burke Road to the intersection with Camberwell Road.

I Drive along Camberwell Road to the intersection with Burwood Road.

J Drive along Burwood Road to the Swinburne University of Technology, which is on the right before the intersection with Glenferrie Road.

6 Technology

Destination data

- Swinburne University of Technology, Map 45 D10

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 12.37 km
- **Total driving time:** 14 min 33 s
- **Average speed:** 51.02 km/h

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

Selected answers (cont'd)

6 Sports and fast cars

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 80 J2 at the centre of Waverley Park, a former AFL football ground.

A Walk due north to Wellington Road.

B Drive along Wellington Road to the intersection with Springvale Road.

C Drive along Springvale Road to the intersection with Princes Highway.

D Drive along Princes Highway to Taco Bill's, just past the intersection with Redpath Crescent.

3 **b** Buy a fast food meal.

E Continue along Princes Highway to Bunnings hardware store, just past the intersection with Smith Road.

4 **c** Shop at a large hardware store

F Continue along Princes Highway to the intersection with Elonera Road.

G Drive along Elonera Road to the first roundabout.

H Drive along Jacksons Road to the intersection with Police Road

5 Two

I Turn left into Police Road.

J Drive along Police Road to the intersection with Princes Highway.

K Drive along Princes Highway to the entrance to Sandown Racecourse.

L Drive along Racecourse Drive to Gate 15, opposite Sandown Road.

Destination data

- Gate 15, Sandown Racecourse, Map 80 C10

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 17.21 km
- **Total driving time:** 16 min 22 s
- **Average speed:** 63.12 km/h

7 Sporting jaunt

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 28 D10 at the centre of the Riverside Golf Driving Range

A Collect car from the corner of Stanford Street and Newsom Street. Drive along Newsom Street to the intersection with Epsom Road.

B Drive along Epsom Road to the intersection with Scotia Street.

C Drive along Scotia Street then Waverley Street to the intersection with Buckley Street.

D Drive along Buckley Street to the intersection with Lincoln Road.

E Drive along Lincoln Road to the intersection with Florence Street.

F Drive along Florence Street and continue into Thistle Street to the intersection with Mt Alexander Road.

G Cross into Brewster Street and turn left into Napier Crescent. Cross the railway line adjacent to Kilmartin Street and turn left into the continuation of Napier Crescent to the intersection with Pascoe Vale Road.

H Drive along Pascoe Vale Road to the Cross Keys Hotel, opposite Strathmore railway station.

3 **d** Have lunch at the Cross Keys Hotel

I Drive south on Pascoe Vale Road, turn right at Napier Crescent and continue to the Brewster Street intersection.

4 Croquet, lawn bowls, indoor swimming and – not too far away – ten pin bowling

J Drive along Napier Street to the intersection with Mt Alexander Road.

K Drive along Mt Alexander Road to the intersection with Dean Street.

L Drive along Dean Street to the intersection with Stuart Street adjacent to the Moonee Valley Racecourse.

Destination data

- Moonee Valley Racecourse, Map 28 K7

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 9.05 km
- **Total driving time:** 10 min 0 s
- **Average speed:** 54.30 km/h

8 Medical mystery tour

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 70 E6 at the St John's Ambulance Headquarters

A Drive along Forster Street to the intersection with Waverley Road.

B Drive along Waverley Road to the intersection with Blackburn Road.

C Drive along Blackburn Road and turn right into Coleman Parade.

D Turn right into Dunscombe Street and stop at the Community Health Centre.

E Drive to the intersection of Blackburn Road and Coleman Parade. Continue on Coleman Parade to Montgomery Avenue, into Lawrence Road, then Wills Avenue, and across an unnamed connecting road to Waimarie Drive. Go east on Waimarie Drive to Stephensons Road and north to Hoskamp Street. Turn left into Hoskamp Street and the Meals on Wheels Centre is a short distance on the left.

F The route to the pizza shop is Hoskamp Street to Avondale Grove, to Miller Crescent, to Alvie Road, to Windsor Avenue, to Hillview Avenue, to Grenfell Road and the Huntingdale Road intersection.

G Drive along Huntingdale Road to the Monash Freeway.

H Enter the Monash Freeway, and exit at Blackburn Road.

J Drive south along Blackburn Road to the intersection with Duerdin Street.

K Drive along Duerdin Street and park outside the former Monash Secondary College.

3 Monash Secondary College

L Drive back to Blackburn Road, on to Wellington Road which becomes North Road, and along Clayton Road to The John Monash Private Hospital.

Destination data

- Sir John Monash Private Hospital, Map 70 C12

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 18.83 km
- **Total driving time:** 16 min 44 s
- **Average speed:** 65.55 km/h

9 Skating round schools

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 18 J1 at the Skateboard Ramp at the J.C. Donath Reserve

A Collect car from Johnson Street and drive south.

B Drive to the first roundabout which intersects Hickford Street.

C Drive along Hickford Street to the intersection with Cheddar Road.

D Drive along Cheddar Road East to the intersection with Cuthbert Road.

E Drive along Cuthbert Road to the skateboard and BMX park.

3 It is a leisure and recreation facility. It provides venues for cycling, BMX, skateboarding, eating and general physical activity.

F Continue along Cuthbert Road to the intersection with Strathmerton Street.

G Drive along Strathmerton Street to the roundabout that intersects with Broadway.

H Drive along Broadway to the intersection with High Street.

I Drive along High Street to the intersection with Queen Street.

J Drive to the two schools via Oakhill Avenue, Ethel Grove, Robb Street and Tyler Street.

K Drive to the Town Hall via Tyler Street and High Street.

L Turn right into Cramer Street.

M Drive along Cramer Street and turn right into Gilbert Road. Continue to the intersection with Henty Street.

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

Selected answers (cont'd)

N Drive to the Guides and Scouts car park via Henty Street, Spratling Street, Griffiths Street and Leamington Street.

Destination data

- The Guides and Scouts car park, Map 18 E4

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 13.82 km
- **Total driving time:** 14 min 14 s
- **Average speed:** 58.28 km/h

10 Off to the skate park

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 45 B2 at the Kew Council Recycling Centre

A Drive along Hutchinson Drive to the intersection with Princess Street.

B Drive along Princess Street to the intersection with Earl Street.

C Drive along Earl Street to the intersection with Willsmere Road.

3 No. Although the Duke St, Peel St, Earl St route is 86 m shorter, the speed imit on these streets is 60 km/h, and therefore this section of the journey would take fractionally longer (0.4 s).

D Drive along Willsmere Road to the intersection with High Street.

E Drive along High Street and park opposite Telstra.

4 **d** Pay a telephone account.

F Continue along High Street to the intersection with Denmark Street.

5 Five food outlets – Yiannis, Di Palma's, Erawan Thai, KFC and McDonalds

G Drive along Denmark Street to the intersection with Barkers Road.

6 The area of the triangle bounded by Barkers Rd, High Street South and Denmark St is 133 472 m^2. The angle at the intersection of High Street South and Barkers Rd is 60.6°.

H Continue along Denmark Street to Mary Street, Christobel Crescent and Ruby Street in which the Hawthorn Football Club Social Club is located.

I The Skate and BMX Park is 2.06 km and 124° east of due north. Proceed south on Ruby Street to Linda Crescent and onto Glenferrie Road. Travel south on Glenferrie Road to Burwood Road, east to Camberwell Road and south-east to Redfern Road which leads directly to the Skate and BMX Park.

Destination data

- The Skate and BMX Park, Map 45 H12

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 10.70 km
- **Total driving time:** 10 min 16 s
- **Average speed:** 61.82 km/h

11 Dented enroute

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 80 D1 at the Waverley City Archery Club

A Collect car from the corner of Monash Drive and Florey Crescent.

B Drive along Monash Drive to the intersection with Wellington Road.

C Drive along Wellington Road and enter the Monash Freeway.

D Exit the freeway at Jacksons Road.

E Drive along Jacksons Road and turn right into Police Road.

F Turn left into Redpath Crescent.

G Drive into the Taco Bill's car park off Princes Highway.

H Drive along Princes Highway to the intersection with Browns Road.

I Drive along Browns Road and park outside the RACV office.

3 To get an assessment of the damage to your car at the RACV Assessment Centre.

J Return along Browns Road to Princes Highway.

K Drive along Princes Highway to the intersection with Corrigan Road.

L Drive along Corrigan Road and turn right into Comber Street. Continue to the intersection with Lawn Road.

M Turn left into Lawn Road and right into Treesbank Avenue. Park at Heatherhill Primary School.

Destination data

- Heatherhill Primary School, Map 80 B12

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 13.92 km
- **Total driving time:** 11 min 15 s
- **Average speed:** 74.32 km/h

12 Sydney sojourn

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 64 R4 at the Glebe Town Hall

A Drive along St Johns Road to the intersection with Jarocin Avenue.

B Drive along Jarocin Avenue and turn right into Bridge Road.

C Continue along Bridge Road and turn left into Parramatta Road.

D Turn right into Australia Street.

E Drive along Australia Street to the intersection with Alton Lane.

F Drive along Alton Lane to the interesection at Denison Street.

G Proceed to Stanmore Public Library via Denison Street, Bedford Street, Liberty Street, Railway Avenue and Douglas Street.

H Continue along Douglas Street and turn right into Percival Road.

I Continue along Percival Street and turn right into Salisbury Street. Proceed to the roundabout and turn left into Northumberland Avenue.

J Continue along Northumberland Avenue and turn right into Parramatta Road. Park by the Bridge Road intersection.

3 McDonalds

K Continue along Parramatta Road and veer left onto Pyrmont Bridge Road.

L Continue along Pyrmont Bridge Road and turn left into Ross Street.

M Veer left into Minogue Street and continue to the roundabout that intersects Nelson Street.

4 Harold Park Paceway, harness racing

N Turn left into Nelson Street and then turn right into Booth Street. Continue to the intersection with Johnston Street.

O Turn left into Johnston Street and park outside the Annandale Community Centre.

Destination data

- Annandale Community Centre, Map 64 L4

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 8.57 km
- **Total driving time:** 8 min 36 s
- **Average speed:** 59.83 km/h

13 From ABC to TV

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 35 A15 at the ABC Enterprises building

A Drive along the private road to the intersection with Campbell Street. Turn right into Reserve Road.

B Turn left into Frederick Street then right into Herbert Street and then left into Ella Street.

C Continue into Dalleys Road and turn right into Berry Avenue.

3 A railway line

4 Tennis

D Turn left into Talus Street and right into Dalleys Road. Cross Willoughby Road and turn into Donnelly Street.

E Turn right into Wheatleigh Street.

F Turn left into the entry road to the Love'n Deuce Tennis Centre

5 Tennis lessons

G Return to Wheatleigh Street and turn right into Chandos Street.

H Continue along Chandos Street to the intersection with Christie Street and turn left. Veer left into Nicholson Street.

I Continue along Nicholson Street and turn left into Rocklands Road, then left onto the Pacific Highway and continue to the intersection with Alexander Street.

© G.A.T.E.WAYS Publications and Blake Education 2010

© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

Selected answers (cont'd)

J Turn right into Alexander Street and continue, turning left into Chandos Street.

K Turn right into Willoughby Road proceed to Artamon Road.

L Turn left into Artamon Road and park at TCN Channel 9 on the left.

Destination data

- Main gate of TCN Channel 9, Map 35 E12

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 7.4 km
- **Total driving time:** 7 min 16 s
- **Average speed:** 61.06 km/h

14 Traffic light trek

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 28 G11 at the main entrance to the Royal Agricultural Show showgrounds.

A Drive along Epsom Road to the intersection with Maribyrnong Road. Cross into Scotia Street.

B Dogleg via Burns Street into Waverley Street and turn left into Corio Street.

C Park at the end of the street.

3 Athletics

D Return to Waverley Street and proceed to the Traffic School via Buckley Street, Tennyson Street and Albion Street.

E Proceed west along Albion Street and Fletcher Street and turn left into Mt Alexander Road.

F Continue to Ormond Road and turn right into Maribyrnong Road.

4 Six

G Turn left into Van Ness Avenue.

5 Maribyrnong River, 100 m wide

H Continue to the roundabout and turn right into Pipemakers Park.

I Continue to the visitors' centre car park.

Destination data

- Pipemakers Park, Map 28 B10

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 13.38 km
- **Total driving time:** 14 min 31 s
- **Average speed:** 55.29 km/h

15 City circuit

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 64 R8 at the Heart Research Centre.

A From the intersection of Carillon Avenue and Missenden Road, drive east on Carillon Avenue to Salisbury Road

B Turn right into Bridge Road and left into Parramatta Road.

C Turn right into Johnston Street.

3 The angle is approximately 63.4°, the distance along Parramatta Road between Bridge Road and Johnston Street is 358 m, therefore the distance to travel on Johnston Street to complete the right-angle triangle is 714 m. The area of the right-angle triangle is approximately 127 800 m^2.

D Continue along Johnston Street and turn left into Booth Street.

E Turn right into Annandale Street and left into Piper Street.

F Turn right into Hutchinson Lane and stop at the tennis courts.

4 Cohen Park Tennis Courts to play a game of tennis

G Continue to Hutchinson Street and turn right. From Hutchinson Street proceed along Annandale Street, Booth Street, Wigram Road, Minogue Crescent and Hereford Street. 'Wattle House Accommodation' is located on the south side of Hereford Street.

5 The journey should take 2 min 19.6 s.

H Return along Hereford Street and turn left into Minogue Crescent.

I Continue to the intersection of Ross Street and Parramatta Road and turn right on to Parramatta Road. Turn left onto Missenden Road.

J Continue to the Heart Research Centre and park.

Destination data

- The Heart Research Centre – your starting point – Map 64 R8

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 7.95 km
- **Total driving time:** 7 min 28 s
- **Average speed:** 63.92 km/h

16 Racing round about

1 Use the map to calculate – will depend on reproduction size of map.

2 **Starting point:** Map 80 J12 at the Melbourne Water Yarraman Creek Retarding Basin

A Exit left onto Elonera Road

B Veer left onto Jacksons Road and continue to the Monash Freeway.

C Enter the Monash Freeway and continue to the Wellington Road exit.

D Exit onto Wellington Road and turn left at Springvale Road.

E Turn left at Mackie Road and stop at the Mulgrave Art Gallery.

F Return to Springvale Road, turn left onto Police Road and continue to the Waverley Gardens Shopping Centre.

G Drive south on Browns Road and stop at the RACV office, just before the Princes Highway.

4 ***b*** To pay an auto club membership fee

H Turn right onto the Princes Highway and continue to the Corrigan Road intersection.

I Turn left onto Corrigan Road and continue to the Allister Road intersection.

J Turn right onto Allister Road and continue to Gate 15 of the Sandown Racecourse and Entertainment Centre.

Destination data

- Gate 15 of the parking area for the Sandown Racecourse and Entertainment Centre, Map 80 B10

Note that answers will be affected by measurement precision and the rounding of calculations.

- **Total distance driven:** 16.37 km
- **Total driving time:** 13 min 44 s
- **Average speed:** 71.55 km/h

© G.A.T.E.WAYS Publications and Blake Education 2010
© Blake Education ISBN 978 1 92571 019 9 Maths in Journeys

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing **the highest quality learning outcomes with instant usability.**

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Maths In Journeys

Do you know people who are bewildered when faced with an urban street directory? *Maths in Journeys* introduces students to the skills they need to navigate street maps effectively.

Students will engage and extend their problem-solving skills as they complete mathematical tasks to follow directions for tracing a journey on the supplied real-life Melways and Sydways maps. Answers are provided to make marking quick and easy.

Topics include

- Navigation skills
- Following directions on a map
- Using map scales
- Using map keys to locate features
- Speed, distance and time calculations
- Calculating average speed
- Using a ruler to measure accurately
- Using a set of compasses to measure accurately
- Solving mathematical, measurement and logic problems

Associated titles in the Instant Lessons series

- Maths In series
  (Event management, Extreme Sports, the Kitchen, the Wild, Fashion, Science, Technology, Building design, Crime, Disasters, Food, Sport, the Community, the Environment, the Workplace, Travel)
- Literacy in Maths (Books 1, 2; Number and Algebra; Data, Measurement and Geometry; A–Z of Maths Terms and Ideas; Maths Across the Curriculum)
- Instant Lessons in Geometry (Books 1, 2)
- Assignments in Junior Measurement (Books 1, 2, 3, 4)
- Assignments in Junior Algebra (Books 1, 2)
- Developing Numeracy series
  - Numbers and the Number System (Books 1, 2, 3)
  - Chance and Data (Books 1, 2, 3)
  - Calculations (Books 1, 2, 3)
  - Algebra (Books 1, 2, 3)
  - Measurement and Space (Books 1, 2, 3)

Visit the Blake Education website or email info@blake.com.au for more information about the Instant Lessons series, other Mathematics titles, the full list of contents for all titles and ordering advice.

Blake EDUCATION
Better ways to learn

Instant Lessons

An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

ISBN: 978-1-92571-019-9